Practice List, Lecture 4: Detailed Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 12 exercises of `Listas/pratica-lecture-04.md`

November 2026

These are detailed solutions to the 12 book exercises of the list `pratica-lecture-04.md` (Lecture 4: credible threats, subgames, subgame-perfect equilibrium, the definition of a repeated game; Stackelberg and time-inconsistency exercises), in the list's order from easy to hard; each heading shows the item's number, level and difficulty from 1 (apply a definition) to 5 (graduate level). Each statement is printed verbatim from the list right before its solution. Every solution has the same six parts: (a) what is asked; (b) the method, with book section, page and lecture slide; (c) the solution, every step shown; (d) the economic interpretation; (e) the verification, by `check_L04.py` in this folder (new items) or the check scripts of `lista-reais-lectures-01-05-solutions/` (reused items), and against the book's worked answer where one exists; (f) common mistakes. Slide numbers in the reused solutions refer to the drafts of the earlier real-exercise list and may differ by a few slides from the final `Aula/Lecture04.pdf`. Items 9 and 12 (time inconsistency, Tadelis §8.3.4) are reading: they are not in the slides of Lecture 4.

Contents

Lecture 4 — Credible threats and subgame perfection **2**

Item 1: Tadelis Exercise 8.8 (p. 171–172) **2**

Item 2: Gibbons Problem 2.12 (p. 135) **5**

Item 3: Gibbons Problem 2.6 (p. 133) **7**

Item 4: Muñoz-Garcia Ch. 4, Exercise 11 (p. 180) **9**

Item 5: Tadelis Exercise 8.13 (p. 173–174) **13**

Item 6: Tadelis Exercise 8.6 (p. 170–171) **16**

Item 7: Tadelis Exercise 8.4 (p. 170) **19**

Item 8: Muñoz-Garcia Ch. 4, Exercise 12 (p. 185–186) **22**

Item 9: Tadelis Exercise 8.15 (p. 174) **24**

Item 10: Tadelis Exercise 8.7 (p. 171) **27**

Item 11: Tadelis Exercise 8.12 (p. 173) **31**

Item 12: Tadelis Exercise 8.16 (p. 174) **34**

Lecture 4 — Credible threats and subgame perfection

Item 1: Tadelis Exercise 8.8

Foundations · difficulty 2/5

Statement (verbatim from the list, item 1)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.8 "Entry Deterrence 1", pp. 171–172 [PDF 190–191]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

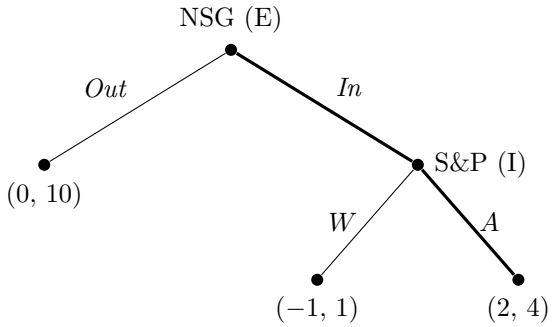
> **Entry Deterrence 1:** NSG Corporation is considering entry into the local phone market in the San Francisco Bay Area. The incumbent, S&P Corporation, predicts that a price war will result if NSG enters. If NSG stays out, S&P earns monopoly profits valued at $10 million (net present value of profits) while NSG earns zero. If NSG enters, it must incur irreversible entry costs of $2 million. If there is a price war, each firm earns $1 million. S&P always has the option of accommodating entry (i.e., not starting a price war). In such a case both firms earn $4 million (net present value). Suppose that the timing is such that NSG first has to choose whether or not to enter the market. Then S&P decides whether to "accommodate entry" or "engage in a price war." What is the subgame-perfect equilibrium outcome to this sequential game? Set up a game tree.

(a) What is asked. Set up the game tree of the entry story (NSG enters or not, then S&P fights or accommodates) and find the subgame-perfect equilibrium outcome.

(b) Method. Backward induction in a finite game of perfect information (Lecture 3, slide 30; Tadelis §8.1, Def. 8.1 and Prop. 8.1, pp. 152–153). With perfect information the subgame-perfect equilibria are exactly the profiles that survive backward induction (Lecture 4, slide 17; Tadelis §8.2, p. 157). Non-credible threats in Nash equilibrium: Lecture 4, slides 8 and 13. Same structure as the entry game of Cabral §7.2 (`Notas/L03/L03-06`).

(c) Solution. **Notation.** Players: NSG (the entrant, E) and S&P (the incumbent, I). E chooses *In* or *Out*; after *In*, I chooses W (price war) or A (accommodate). Payoffs (u_E, u_I) in $ million of net present value. The entry cost of 2 is sunk once E enters, so it is subtracted from every payoff that follows *In*:

$$\begin{aligned}(u_E, u_I)(Out) &= (0,\ 10),\\ (u_E, u_I)(In, W) &= (1-2,\ 1) = (-1,\ 1),\\ (u_E, u_I)(In, A) &= (4-2,\ 4) = (2,\ 4).\end{aligned}$$



Payoffs (u_E, u_I); thick branches = backward-induction choices.

Step 1: the incumbent's node. After entry, S&P compares

$$u_I(In, A) = 4 > 1 = u_I(In, W),$$

so it accommodates. The price war is not a credible response.

Step 2: the entrant's node. Replacing S&P's node by its outcome $(2, 4)$, NSG compares

$$u_E(In) = 2 > 0 = u_E(Out),$$

so it enters. SPE: (In, A), outcome entry and accommodation, payoffs $(2, 4)$. The game has two proper subgames (the whole game and the one at S&P's node), and the SPE plays a Nash equilibrium in both.

Step 3: all Nash equilibria. Normal form (rows: E's strategies; columns: I's strategies):

	W	A
Out	$\underline{0},\ \underline{10}$	$0,\ \underline{10}$
In	$-1,\ 1$	$\underline{2},\ \underline{4}$

Pure Nash equilibria: (In, A) and (Out, W). With $w = \Pr_I(W)$, NSG's payoff from entering is $-w + 2(1 - w) = 2 - 3w$, so Out is a best reply iff $0 \geq 2 - 3w$, i.e. $w \geq \frac{2}{3}$; S&P is not reached and is indifferent. The full set is

$$\{(In, A)\} \ \cup\ \{(Out,\ w\,W + (1 - w)A) : w \geq \tfrac{2}{3}\}.$$

Every equilibrium in the second family fails sequential rationality at the node S&P never reaches (Tadelis Def. 8.1, p. 152): the threat "war with probability at least $\frac{2}{3}$" costs nothing because it is never carried out, and it would not be carried out $(1 < 4)$.

Step 4: what would make the threat credible (deepening). Suppose S&P could, before NSG decides, commit to a penalty b that it pays if it accommodates (a public contract, a capacity investment, a "most-favored-customer" clause). Then at its node S&P fights iff $1 > 4 - b$, i.e. $b > 3$; NSG, anticipating war, stays out $(-1 < 0)$, and S&P earns 10 without ever paying b. A credible commitment is worth up to $10 - 4 = 6$ to S&P. Independently, an entry cost above 4 would deter entry even with accommodation $(4 - \text{cost} < 0)$.

(d) Interpretation. S&P's prediction of a price war is cheap talk: once NSG has sunk its \$2 million, a war gives S&P 1 instead of 4, and NSG knows it. These are the three lessons of the Brexit case (Lecture 4, slide 11) in miniature: *talk is cheap* (announcing war without changing payoffs is not credible); *sunk investments can be strategic* (only a sunk, payoff-changing move such as the penalty b makes war credible); *observability matters* (the commitment works only if NSG sees it before deciding). The outcome does not depend on how the entry cost is booked: if "each firm earns \$1 million" were already net of it, NSG would still enter $(2 > 0$, and even a war would pay $1 > 0)$.

(e) Verification. `check_B_L04.py` recomputes both backward-induction steps, the two pure Nash equilibria of the 2×2 normal form, scans w on a grid of step $\frac{1}{300}$ (Nash equilibria exactly at $w = 0$ with entry and $w \geq \frac{2}{3}$ with *Out*), and checks the commitment thresholds $b > 3$ and the value 6. Output line: `tadelis-8-8: SPE (In, A); NE (Out, Pr(W) >= 2/3); commitment worth up to 6`. Tadelis prints no answer.

(f) Common mistakes.

- Forgetting to subtract the sunk entry cost from NSG's payoffs after entry.
- Reporting (Out, W) as an SPE because it is a Nash equilibrium.
- Giving the outcome only ("NSG enters") when the SPE is a strategy profile: S&P's plan A must be stated.

Answer. Tree above. Subgame-perfect equilibrium: NSG plays *In*, S&P plays A (accommodate); outcome: entry and accommodation, payoffs $(2, 4)$ in \$ million (NSG net of the entry cost). The other Nash equilibria, *Out* with $\Pr(W) \geq \frac{2}{3}$, rest on a non-credible threat.

Item 2: Gibbons Problem 2.12

Foundations · difficulty 1/5

Statement (verbatim from the list, item 2)

Source. Gibbons (1992), *Game Theory for Applied Economists*, §2.3 Problems, Problem 2.12, p. 135 [PDF 142]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 2.12. What is a strategy in a repeated game? What is a subgame in a repeated game? What is a subgame-perfect Nash equilibrium?

(a) What is asked. Define a strategy, a subgame and a subgame-perfect Nash equilibrium in a repeated game.

(b) Method. The definitions of Gibbons §2.3.B (pp. 93–95), stated for the finitely repeated game $G(T)$ and the infinitely repeated game $G(\infty,\delta)$; the matching definitions in the slides (Lecture 4, slides 21–22: repeated game, $G(T,\delta)$, "a strategy says what to play in each period after each possible history", "each period starts a new subgame"; SPE, slide 17) and in Tadelis §9.3–9.4 (pp. 178–182). Each definition is illustrated with the two-stage Prisoners' Dilemma of Gibbons Fig. 2.3.1 (p. 82).

(c) Solution. Setting. A stage game $G = \{A_1,\dots,A_n;\ u_1,\dots,u_n\}$ is played in periods $t = 1,2,\dots$; before each period all players observe all past actions. The *history* through period t is $h^t = (a^1,\dots,a^t)$, with $a^s = (a_1^s,\dots,a_n^s)$ the action profile of period s; $h^0 = \varnothing$. H^t is the set of histories through t.

1. Strategy. A strategy of player i specifies the action he will take in each stage, for each possible history of play through the previous stage (Gibbons p. 94):

$$s_i = (s_i^1, s_i^2, \dots), \qquad s_i^t : H^{t-1} \to A_i.$$

It is a complete contingent plan, including histories that the plan itself would never produce. *Example.* In the two-stage Prisoners' Dilemma (actions L_i, R_i) there are four first-stage outcomes, so a strategy has five instructions (v,w,x,y,z): a first-stage action and a second-stage action after each of (L_1,L_2), (L_1,R_2), (R_1,L_2), (R_1,R_2) (Gibbons p. 93). Each player has $2^5 = 32$ pure strategies. In general, with a 2×2 stage game played T times, player i has $1+4+\cdots+4^{T-1} = \frac{4^T-1}{3}$ information sets and $2^{(4^T-1)/3}$ pure strategies.

2. Subgame. A subgame is the piece of the game that remains to be played from any point at which the complete history so far is common knowledge (Gibbons p. 94). In $G(T)$ a subgame beginning at stage $t+1$ is the repeated game in which G is played $T-t$ times, $G(T-t)$, following one particular history h^t; there is one such subgame for each $h^t \in H^t$. In $G(\infty,\delta)$ each subgame beginning at stage $t+1$ is identical to $G(\infty,\delta)$ itself, and again there is one for each history through t. *Not* a subgame: stage t taken on its own, for $t < T$; a subgame includes all the moves that follow its starting point (Gibbons p. 95). *Example.* The two-stage Prisoners' Dilemma has four proper subgames after period 1 (one per first-stage outcome), each a one-shot Prisoners' Dilemma, plus the whole game.

3. Subgame-perfect Nash equilibrium. A Nash equilibrium is subgame-perfect if the players' strategies constitute a Nash equilibrium in every subgame (Selten 1965; Gibbons p. 95; Lecture 4, slide 17 states it for general extensive forms: the restriction of the profile to every proper subgame

is a Nash equilibrium of that subgame). In a repeated game this means: after *every* history h^t, on or off the equilibrium path, the continuation strategies $s|_{h^t}$ form a Nash equilibrium of the continuation game.

Example (computed). In the two-stage Prisoners' Dilemma ($u(L,L) = (1,1)$, $u(L,R) = (5,0)$, $u(R,L) = (0,5)$, $u(R,R) = (4,4)$, payoffs summed over the two stages), enumerating all 32×32 profiles gives 16 Nash equilibria; all of them have the outcome (L_1, L_2) in both stages, payoffs $(2,2)$, but they differ in what is planned after unreached first-stage outcomes. Exactly one is subgame perfect: each player plays L in stage 1 and L after every history, (L, L, L, L, L), because each of the four second-stage subgames is a one-shot Prisoners' Dilemma whose only Nash equilibrium is (L_1, L_2).

(d) Interpretation. Three ideas carry over from Lectures 3–4 to repeated games. A strategy must say what happens after deviations, because those plans are what make promises and threats; a subgame starts after each observed history; and subgame perfection requires the punishments and rewards to be ones players would actually carry out. This is the toolkit for the carrots-and-sticks and folk-theorem arguments of Lecture 5: cooperation can be sustained only by continuation play that is itself an equilibrium.

(e) Verification. `check_B_L04.py` enumerates the 32 strategies (v, w, x, y, z) of each player in the two-stage Prisoners' Dilemma, computes all Nash equilibria of the 32×32 game and keeps those that play a stage Nash equilibrium in every second-stage subgame; it also checks the count $\frac{4^T-1}{3}$ for $T = 1, 2, 3$ and the 3^{10} strategies of the 3×3 two-stage game of Gibbons Fig. 2.3.3. Output line: `gibbons-2-12: 32 strategies each, 16 NE profiles, 1 SPE (L,L,L,L,L) each`. Gibbons prints no separate answer; the definitions are those of pp. 94–95.

(f) Common mistakes.

- Defining a strategy as a sequence of actions, one per period ("play b, then c"): it must be contingent on the history.
- Calling the stage game played in period t a subgame: a subgame includes all later periods.
- Checking Nash behavior only on the equilibrium path: subgame perfection is about every history, including those after deviations.

Answer. A strategy assigns an action in every period to every possible history of play through the previous period. A subgame is the remainder of the repeated game after a given history (in $G(T)$, the game $G(T-t)$ after one history h^t; in $G(\infty, \delta)$, a copy of the whole game). A subgame-perfect Nash equilibrium is a strategy profile whose restriction to every subgame, after every history, is a Nash equilibrium of that subgame.

Item 3: Gibbons Problem 2.6

Foundations · difficulty 3/5

Statement (verbatim from the list, item 3)

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 Problems (Section 2.2), Problem 2.6, p. 133 [PDF 140]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 2.6. Three oligopolists operate in a market with inverse demand given by $P(Q) = a - Q$, where $Q = q_1 + q_2 + q_3$ and q_i is the quantity produced by firm i. Each firm has a constant marginal cost of production, c, and no fixed cost. The firms choose their quantities as follows: (1) firm 1 chooses $q_1 \geq 0$; (2) firms 2 and 3 observe q_1 and then simultaneously choose q_2 and q_3, respectively. What is the subgame-perfect outcome?

(a) What is asked. Three firms face $P(Q) = a - Q$ with marginal cost c. Firm 1 chooses q_1; firms 2 and 3 observe it and then choose q_2, q_3 simultaneously. What is the subgame-perfect outcome?

(b) Method. Two-stage game with a simultaneous second stage (Gibbons §2.2.A, pp. 71–73): for each q_1 solve the followers' Cournot game (a Nash equilibrium of the subgame that follows q_1), then let firm 1 maximize against it. The outcome is a backward-induction outcome (Lecture 4, slides 31–33 on SPE; Tadelis §8.3.2, pp. 160–163). Tadelis 8.4 (item 7) poses the same game and asks for the strategies, the number of subgames and a non-SPE Nash equilibrium too; this item stops at the outcome.

(c) Solution. Assume $a > c$ and $q_1 \leq a - c$ (above $a - c$ the price is below c and firm 1 loses money).

Step 1: the followers' subgame after q_1. Firm $i \in \{2, 3\}$ solves $\max_{q_i \geq 0} (a - q_1 - q_i - q_j - c)\, q_i$, with first-order condition $a - c - q_1 - 2q_i - q_j = 0$, so $BR_i(q_j) = \frac{a-c-q_1-q_j}{2}$. The two best replies have slope $-\frac{1}{2}$ and cross once; by symmetry the unique Nash equilibrium of the subgame is

$$q_2 = q_3 = \frac{a - c - q_1}{3}.$$

(Check: $\frac{a-c-q_1-q_j}{2} = \frac{a-c-q_1}{2} \cdot \frac{2}{3} = \frac{a-c-q_1}{3}$.)

Step 2: firm 1. Substituting the followers' total output $\frac{2(a-c-q_1)}{3}$, the price is $P = a - q_1 - \frac{2(a-c-q_1)}{3}$, so

$$\pi_1(q_1) = \Big(\frac{a - c}{3} - \frac{q_1}{3}\Big) q_1 = \frac{(a - c - q_1)\, q_1}{3}.$$

It is strictly concave; the first-order condition $a - c - 2q_1 = 0$ gives $q_1^* = \frac{a-c}{2}$.

Outcome.

$$q_1^* = \frac{a - c}{2}, \qquad q_2^* = q_3^* = \frac{a - c}{6}, \qquad Q^* = \frac{5(a - c)}{6}, \qquad P^* = \frac{a + 5c}{6},$$

with profits $\pi_1 = \frac{(a-c)^2}{12}$ and $\pi_2 = \pi_3 = \frac{(a-c)^2}{36}$. Compare the three-firm Cournot outcome $\frac{a-c}{4}$ each, $Q = \frac{3(a-c)}{4}$, $\pi_i = \frac{(a-c)^2}{16}$: the leader produces twice the Cournot quantity, earns $\frac{4}{3}$ of its Cournot profit, and each follower earns $\frac{4}{9}$ of its Cournot profit.

(d) Interpretation. Moving first lets firm 1 commit to a large output; the followers, each taking q_1 as given, shrink their own output (a unit more from firm 1 reduces each follower by $\frac{1}{3}$, not $\frac{1}{2}$ as in a duopoly), so the leader's expansion crowds out only $\frac{2}{3}$ of a unit. That is why the leader overproduces relative to Cournot, and why the followers are worse off.

(e) Verification. `check_L04.py` solves the followers' subgame and the leader's problem with sympy (symbolic in a, c), confirms the quantities, the price and the three profits, and compares with the three-firm Cournot profits ($\frac{4}{3}$ and $\frac{4}{9}$), and checks with exact fractions on a grid of q_1 that $\frac{a-c}{2}$ maximizes π_1 for $a = 100$, $c = 10$. Output line: `gibbons-2-6: q = ((a-c)/2, (a-c)/6, (a-c)/6), P = (a+5c)/6`. Gibbons prints no answer.

(f) Common mistakes.

- Solving a two-firm Stackelberg game with firm 1 against "the market": the followers play a *Cournot* game with each other after q_1, so the leader faces $\frac{2(a-c-q_1)}{3}$, not $\frac{a-c-q_1}{2}$.
- Reporting only q_1^*: the outcome is the whole vector (q_1, q_2, q_3).

Answer. $q_1^* = \frac{a-c}{2}$, $q_2^* = q_3^* = \frac{a-c}{6}$, price $P^* = \frac{a+5c}{6}$; profits $\frac{(a-c)^2}{12}$ for the leader and $\frac{(a-c)^2}{36}$ for each follower.

Item 4: Muñoz-Garcia Ch. 4, Exercise 11

Exam · difficulty 3/5

Statement (verbatim from the list, item 4)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 4, Exercise 11 "Stackelberg with Two Firms", p. 180 [PDF 192]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider a leader and a follower in a Stackelberg game of quantity competition. Firms face an inverse demand curve $p(Q) = 1 - Q$, where $Q = q_L + q_F$ denotes aggregate output. The leader faces a constant marginal cost $c_L > 0$ while the follower's constant marginal cost is $c_F > 0$, where $1 > c_F \geq c_L$, indicating that the leader has access to a more efficient technology than the follower.
>
> **Part (a)** Find the follower's best response function, BRF_F, i.e., $q_F(q_L)$.
>
> **Part (b)** Determine each firm's output strategy in the subgame perfect equilibrium (SPNE) of this sequential-move game.
>
> **Part (c)** Under which conditions on c_L can you guarantee that both firms produce strictly positive output levels?
>
> **Part (d)** Assuming cost symmetry, i.e., $c_L = c_F = c$, determine the aggregate equilibrium output level in the Stackelberg game, and its associated equilibrium price. Then compare them with those arising in (1) a monopoly; in (2) the Cournot game of quantity competition; and (3) in the Bertrand game of price competition (which, under cost symmetry, is analogous to a perfectly competitive industry) [*Hint*: For a more direct and visual comparison, depict the inverse demand curve $p(Q) = 1 - Q$, and locate each of these four aggregate output levels in the horizontal axis (with its associated prices in the vertical axis).].

(a) What is asked. Stackelberg duopoly with $p = 1 - Q$ and costs $c_L \leq c_F < 1$: (a) the follower's best response; (b) the SPNE strategies; (c) when both outputs are positive; (d) with $c_L = c_F = c$, total output and price, compared with monopoly, Cournot and Bertrand.

(b) Method. Backward induction in a perfect-information game with a continuum of actions: solve the follower's problem for *every* q_L, substitute into the leader's problem (Tadelis §8.3.2, "Stackelberg Competition", pp. 160–163; SPE Def. 8.3, p. 157). The Cournot benchmark is Tadelis §5.2.3, p. 87. Lecture 4: SPE definition and the backward-induction fact, slide 17; sunk, observable commitments, slide 11.

(c) Solution. **(a) Follower.** Given q_L, firm F solves $\max_{q_F \geq 0} (1 - q_L - q_F)q_F - c_F q_F$. The objective is strictly concave (second derivative -2), and the first-order condition $1 - q_L - 2q_F - c_F = 0$ gives

$$q_F(q_L) \;=\; \max\Big\{\frac{1 - c_F - q_L}{2},\; 0\Big\} \;=\; \frac{1 - c_F}{2} - \frac{q_L}{2} \quad \text{for } q_L \leq 1 - c_F. \tag{1}$$

The best response falls with the leader's output (slope $-\frac{1}{2}$) and shifts down when c_F rises. For $q_L \geq 1 - c_F$ the residual demand no longer covers the follower's cost, and it produces nothing.

(b) Leader. On $q_L \leq 1 - c_F$, substituting (1), the price is $1 - q_L - \frac{1 - c_F - q_L}{2} = \frac{1 + c_F}{2} - \frac{q_L}{2}$, and the leader solves

$$\max_{q_L} \Big(\frac{1 + c_F}{2} - \frac{q_L}{2} - c_L\Big) q_L \quad \Longrightarrow \quad \frac{1 + c_F}{2} - q_L - c_L = 0 \quad \Longrightarrow \quad q_L^* = \frac{1 + c_F - 2c_L}{2}.$$

The objective is strictly concave (second derivative -1). Then

$$q_F^* = \frac{1-c_F}{2} - \frac{1+c_F-2c_L}{4} = \frac{1-3c_F+2c_L}{4}, \qquad p^* = 1-q_L^*-q_F^* = \frac{1+c_F+2c_L}{4}.$$

The SPNE is the *pair of strategies*

$$\Big(q_L^* = \tfrac{1+c_F-2c_L}{2},\ \ q_F(q_L) = \max\{\tfrac{1-c_F-q_L}{2}, 0\} \text{ for every } q_L \geq 0\Big),$$

not just the two numbers: the follower's plan covers every output the leader could choose, on and off the path. Each firm's output falls with its own cost and rises with the rival's.

When the formula applies. $q_L^* \leq 1-c_F$ iff $c_L \geq \frac{3c_F-1}{2}$, i.e. exactly when $q_F^* \geq 0$. Otherwise the leader's optimum is not interior and the follower is out of the market. For $q_L \geq 1-c_F$ the leader is a monopolist with profit $(1-q_L-c_L)q_L$, maximized at $\frac{1-c_L}{2}$. Then:

- $c_L > \frac{3c_F-1}{2}$: interior, as above, both firms produce;
- $2c_F - 1 < c_L \leq \frac{3c_F-1}{2}$: *limit output*, $q_L^* = 1-c_F$, $q_F^* = 0$ (the leader's profit rises up to $1-c_F$ on the left piece and falls beyond it on the right piece, because $\frac{1-c_L}{2} < 1-c_F$);
- $c_L \leq 2c_F - 1$: *blockaded entry*, $q_L^* = \frac{1-c_L}{2}$ (monopoly), $q_F^* = 0$.

For example $(c_L, c_F) = (\frac{1}{4}, \frac{1}{2})$ gives limit output $q_L^* = \frac{1}{2}$, and $(\frac{1}{10}, \frac{7}{10})$ gives blockaded entry $q_L^* = \frac{9}{20}$.

(c) Both outputs positive.

$$q_F^* > 0 \iff 1-3c_F+2c_L > 0 \iff c_L > \frac{3c_F-1}{2}.$$

The leader's condition $q_L^* > 0 \iff c_L < \frac{1+c_F}{2}$ never binds: with $c_L \leq c_F < 1$, $q_L^* \geq \frac{1-c_F}{2} > 0$. So both firms produce iff

$$\frac{3c_F-1}{2} < c_L \leq c_F,$$

a non-empty band because $\frac{3c_F-1}{2} < c_F \iff c_F < 1$. If $c_F \leq \frac{1}{3}$ the lower bound is not positive, so every $c_L \in (0, c_F]$ works. The leader's cost advantage must not be too large.

(d) Symmetric costs, $c_L = c_F = c$. The condition in (c) holds ($c > \frac{3c-1}{2}$), so the interior formulas apply:

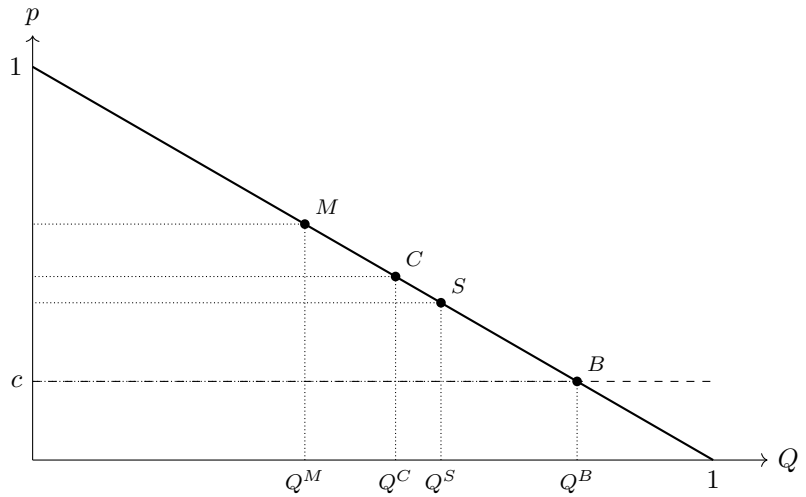
$$q_L^* = \frac{1-c}{2}, \qquad q_F^* = \frac{1-c}{4}, \qquad Q^S = \frac{3(1-c)}{4}, \qquad p^S = 1-Q^S = \frac{1+3c}{4}.$$

The follower produces half of the leader's output. Benchmarks:

- *Monopoly*: $\max_Q (1-Q-c)Q$ gives $Q^M = \frac{1-c}{2}$, $p^M = \frac{1+c}{2}$.
- *Cournot*: $q_i = \frac{1-c-q_j}{2}$ for both firms gives $q_i = \frac{1-c}{3}$, $Q^C = \frac{2(1-c)}{3}$, $p^C = \frac{1+2c}{3}$.
- *Bertrand* (= perfect competition with equal costs): $p^B = c$, $Q^B = 1-c$.

Because $\frac{1}{2} < \frac{2}{3} < \frac{3}{4} < 1$,

$$Q^M < Q^C < Q^S < Q^B \qquad \text{and} \qquad p^M > p^C > p^S > p^B.$$



Demand $p = 1 - Q$ drawn for $c = 0.2$: $Q^M = 0.4$, $Q^C \approx 0.53$, $Q^S = 0.6$, $Q^B = 0.8$; prices $\frac{1+c}{2}$, $\frac{1+2c}{3}$, $\frac{1+3c}{4}$, c.

Profits: the leader earns $(p^S - c)q_L^* = \frac{(1-c)^2}{8}$, more than its Cournot profit $\frac{(1-c)^2}{9}$; the follower earns $\frac{(1-c)^2}{16}$, less.

(d) Interpretation. The leader's advantage comes from commitment, not from information: its output is observed and cannot be revised, so the follower takes it as given and contracts. The leader expands past its Cournot output ($\frac{1-c}{2} > \frac{1-c}{3}$) because it knows the follower will respond by cutting back. The follower would like to threaten "I produce $\frac{1-c}{3}$ whatever you do". Against that plan, $q_L = \frac{1-c}{3}$ is a best reply, so the Cournot outcome is a Nash equilibrium of the sequential game (Tadelis p. 162). The threat is not credible: once q_L is sunk, $\frac{1-c}{3}$ is not the follower's best response to any $q_L \neq \frac{1-c}{3}$. Subgame perfection removes it (slide 17). The cost regimes in (b) carry the same idea: a large enough cost advantage lets the leader commit to the output $1 - c_F$, which leaves the follower no room to produce (limit output). This is the quantity version of entry deterrence through an observable, irreversible commitment (slide 11). For consumers, sequential quantity competition is more competitive than Cournot: total output rises from $\frac{2}{3}$ to $\frac{3}{4}$ of the competitive level.

(e) Verification. `check_D.py` derives BR_F, q_L^*, q_F^*, p^* and the cutoff $c_L = \frac{3c_F - 1}{2}$ with sympy. For six cost pairs covering the three regimes it finds the leader's best output by exact search over $q_L \in \{k/2000\}$, using the truncated best response, and confirms the regime formula each time. It also checks the four symmetric totals, prices and profits, the strict ranking at four values of c, and that the Cournot outcome is a Nash but not a subgame-perfect profile. Output: "Munoz-Garcia 4.11: BR_F = (1-cF-qL)/2; qL* = (1+cF-2cL)/2, qF* = (1-3cF+2cL)/4; both > 0 iff cL > (3cF-1)/2; Q: monopoly (1-c)/2 < Cournot 2(1-c)/3 < Stackelberg 3(1-c)/4 < Bertrand 1-c (book: same)". *Comparison with Muñoz-Garcia's answer* (pp. 181–185): the same best response, SPNE, cutoffs (the book's C_A: $c_L > -\frac{1}{2} + \frac{3}{2}c_F$; C_B: $c_L < \frac{1}{2} + \frac{1}{2}c_F$, not binding below the 45° line), symmetric quantities and the ranking in its Fig. 4.34. The book omits the truncation at $q_F = 0$, so its best response and SPNE formula are valid only inside the region of part (c). The limit-output and blockaded regimes above are not in the book.

(f) Common mistakes.

- Reporting the SPNE as the numbers (q_L^*, q_F^*). The follower's strategy is the whole function $q_F(q_L)$; the numbers are the equilibrium outcome.

- Solving both first-order conditions simultaneously, which gives the Cournot quantities $\frac{1-c}{3}$, instead of substituting the follower's response into the leader's profit first.
- Using the interior formula $q_F^* = \frac{1-3c_F+2c_L}{4}$ when it is negative. When it is, the follower produces zero and the leader's problem changes.

Answer. (a) $q_F(q_L) = \max\{\frac{1-c_F-q_L}{2}, 0\}$. (b) SPNE: $q_L^* = \frac{1+c_F-2c_L}{2}$ and the follower plays $q_F(q_L)$ after every q_L; outcome $q_F^* = \frac{1-3c_F+2c_L}{4}$, $p^* = \frac{1+c_F+2c_L}{4}$ (valid when $q_F^* \geq 0$). (c) Both produce iff $c_L > \frac{3c_F-1}{2}$; $q_L^* > 0$ always. (d) $Q^S = \frac{3(1-c)}{4}$, $p^S = \frac{1+3c}{4}$; $Q^M = \frac{1-c}{2} < Q^C = \frac{2(1-c)}{3} < Q^S < Q^B = 1-c$, and prices in the reverse order (book: same).

Item 5: Tadelis Exercise 8.13

Exam · difficulty 4/5

Statement (verbatim from the list, item 5)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.13 "Junk Mail Advertising", pp. 173–174 [PDF 192–193]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Junk Mail Advertising:** Suppose a single good is owned by a single seller who values it at $c > 0$ (he can consume the good and get a payoff of c). There is a single buyer who has a small transportation cost $k > 0$ to get to and back from the seller's store, and he values the good at $v > c + k$. The buyer first decides whether to make the commute or stay at home, not buy the good, and receive a payoff of 0. If the buyer commutes to the store, the seller can then make the buyer a take-it-or-leave-it price offer $p \geq 0$. The buyer can then accept the offer, pay p, and get the good, or he can walk out and not buy the good. Assume that c, v, and k are common knowledge.
>
> a. As best you can, draw the extensive form of this game. What is the best response of the buyer at the node at which he decides whether to accept or reject the seller's offer?
>
> b. Find the subgame-perfect equilibrium of the game and show that it is unique. Is it Pareto optimal?
>
> c. Are there Nash equilibria that yield a higher payoff to both players as compared to the subgame-perfect equilibrium you found in (b)?
>
> d. Now assume that before the game is played the seller can, at a small cost $\varepsilon < (v - c - k)$, send the buyer a postcard that commits the seller to a certain price at which the buyer can buy the good (e.g., "bring this coupon and get the good at a price p"). Would the seller choose to do so? Justify your answer with an equilibrium analysis.

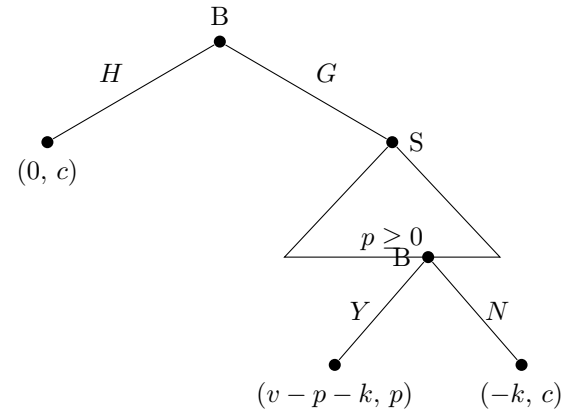
(a) What is asked. A buyer pays a commuting cost k to reach a seller, who then names a take-it-or-leave-it price. (a) Extensive form and the buyer's reply at the acceptance node; (b) the SPE, its uniqueness and Pareto optimality; (c) Nash equilibria better for both; (d) whether the seller would pay ε to mail a postcard that commits him to a price.

(b) Method. Backward induction with a continuum of actions (Lecture 3, slides 30 and 37; Tadelis §8.1 and §8.3.2, the Stackelberg logic, pp. 160–163); the ultimatum-game argument for ties (Tadelis Prop. 11.2, p. 223: the responder must accept at indifference, or the proposer has no best offer). Nash equilibria sustained by off-path threats (Lecture 4, slides 8 and 13). Commitment as a move added before the game (Lecture 4, slide 11; `Notas/L03/L03-06`, Cabral §7.2).

(c) Solution. Notation. Buyer (B) and seller (S). B chooses G (go to the store) or H (stay home); then S names $p \geq 0$; then B accepts (Y) or walks out (N). Data: $c > 0$, $k > 0$, $v > c + k$. Payoffs (u_B, u_S):

$$H: \ (0, \ c); \qquad G, p, Y: \ (v - p - k, \ p); \qquad G, p, N: \ (-k, \ c).$$

(a) Extensive form and the buyer's reply.



The triangle stands for the continuum of prices; one typical price p is drawn.

At the node after price p, the commuting cost is already sunk; it appears in both payoffs. B accepts if

$$v - p - k \;\geq\; -k \iff p \leq v.$$

So the best reply is: accept every $p < v$, reject every $p > v$, and either action at $p = v$.

(b) SPE, uniqueness, efficiency.

Step 1: the acceptance rule at $p = v$. Suppose B rejected $p = v$. Then S would get p for every $p < v$ and $c < v$ otherwise, so his payoff approaches v as $p \uparrow v$ but never reaches it; S would have no best offer, so the profile cannot be an SPE. Hence in every SPE B accepts if and only if $p \leq v$.

Step 2: the seller. Given that rule, S's payoff is p for $p \leq v$ and $c < v$ for $p > v$; the unique maximizer is $p^* = v$.

Step 3: the buyer's commuting decision. Going yields

$$u_B(G) = v - p^* - k = v - v - k = -k \;<\; 0 = u_B(H),$$

so B stays home. Every step had a unique optimum, so the SPE is unique:

$$\text{B: } H\text{, and accept iff } p \leq v; \qquad \text{S: offer } p = v. \qquad \text{Outcome: no trade, payoffs } (0,\ c).$$

Not Pareto optimal. Trade at any price $p \in (c,\ v - k)$ (nonempty since $v > c + k$) gives $u_B = v - p - k > 0$ and $u_S = p > c$: both strictly better. The lost surplus is $v - c - k > 0$. This is a hold-up: the commuting cost is sunk when the price is set, so S extracts all of v and B, anticipating it, never makes the trip.

(c) Better Nash equilibria. Yes. Fix $\hat{p} \in (c,\ v - k)$ and consider

$$\text{B: } G\text{, accept iff } p \leq \hat{p}; \qquad \text{S: offer } \hat{p}.$$

S's best reply: offers $p \leq \hat{p}$ are accepted and give $p \leq \hat{p}$; offers above are rejected and give $c < \hat{p}$. So $\hat{p}$ is optimal. *B's best reply:* given $\hat{p}$, going and accepting gives $v - \hat{p} - k > 0$, more than staying (0) or going and rejecting $(-k)$; B's rule at prices other than $\hat{p}$ is off the path and does not affect his payoff. Payoffs $(v - \hat{p} - k,\ \hat{p})$ beat $(0, c)$ for both. The same construction is a Nash equilibrium for every $\hat{p} \in [c, v - k]$ (the endpoints give one player only a weak gain), so the set of Nash payoffs that Pareto-dominate the SPE is $\{(v - \hat{p} - k, \hat{p}) : \hat{p} \in (c, v - k)\}$. None is subgame perfect: after an off-path offer $p \in (\hat{p}, v)$, rejecting gives $-k < v - p - k$, so the threat to walk out is not credible.

(d) The postcard. Add a first move: S either mails a postcard committing to $\hat{p}$ (at cost ε) or does not. Without the postcard the game is the one in (b): payoff c to S. With it, B buys at $\hat{p}$ if

he goes and $\hat{p} \le v$, so he goes iff $v - \hat{p} - k \ge 0$, i.e. $\hat{p} \le v - k$. By the argument of Step 1, B goes at indifference. Then S's payoff from the postcard is

$$u_S(\hat{p}) = \begin{cases} \hat{p} - \varepsilon, & \hat{p} \le v - k, \\ c - \varepsilon, & \hat{p} > v - k, \end{cases} \qquad \Rightarrow \qquad \hat{p}^* = v - k, \quad u_S = v - k - \varepsilon.$$

Compared with not mailing,

$$(v - k - \varepsilon) - c = (v - c - k) - \varepsilon > 0 \quad \text{since } \varepsilon < v - c - k.$$

So in the SPE the seller mails the postcard with price $v - k$; B goes and buys; payoffs $(0,\ v - k - \varepsilon)$. Trade now happens; the total surplus is $v - c - k - \varepsilon$, all of it to the seller.

(d) Interpretation. The commitment in (c) has to come from the buyer and is not credible; the commitment in (d) comes from the seller, who limits his own future options, and it is credible because the coupon binds him. These are the three Lecture 4 lessons (slide 11) in one example: the buyer's *sunk* commuting cost is what exposes him to hold-up; "I will only pay $\hat{p}$" is *cheap talk*; and the postcard works because it is *observable* before the buyer decides and changes what the seller can do afterwards. "Junk mail" has value even when it contains no information. The seller still captures the whole surplus; the coupon only removes the inefficiency, the same logic as a bond in the entry game (Cabral §7.2).

(e) Verification. `check_B_L04.py` uses $c = 2$, $v = 10$, $k = 1$, $\varepsilon = \frac{1}{2}$ and prices on a 0.25 grid: the seller's best offer is $p = 10 = v$ and the buyer stays home; under "reject at v" the best grid offer is $v - 0.25$ (no maximum on a continuum); thresholds $\hat{p} \in \{2.5, 5, 8.5\}$ give Nash equilibria that beat $(0, c)$ for both; with the postcard, $\hat{p}^* = 9 = v - k$ and $8.5 > 2$; sympy confirms the gain $(v - c - k) - \varepsilon$. Output line: `tadelis-8-13: SPE p = v, no trade; postcard at v - k is bought, seller nets v - k - eps`. Tadelis prints no answer.

(f) Common mistakes.

- Letting the buyer reject at $p = v$ in the SPE: then the seller has no optimal offer, so no SPE exists with that rule.
- Including k in the acceptance decision ("accept iff $p \le v - k$"): k is sunk at that node.
- Pricing the postcard at v: the buyer would then not commute; the coupon must leave him $v - \hat{p} - k \ge 0$.

Answer. (a) Tree above; at the acceptance node the buyer accepts iff $p \le v$ (the sunk k does not matter). (b) Unique SPE: seller offers $p = v$, buyer accepts iff $p \le v$ and therefore stays home; payoffs $(0, c)$; not Pareto optimal, since trade at any $p \in (c, v - k)$ helps both. (c) Yes: buyer goes and accepts only $p \le \hat{p}$, seller offers $\hat{p}$, for any $\hat{p} \in (c, v - k)$; Nash but not subgame perfect. (d) Yes: the seller mails a coupon at $\hat{p} = v - k$, the buyer comes and buys, seller earns $v - k - \varepsilon > c$.

Item 6: Tadelis Exercise 8.6

Exam · difficulty 3/5

Statement (verbatim from the list, item 6)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.6 "Investment in the Future", pp. 170–171 [PDF 189–190]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Investment in the Future:** Consider two firms that play a Cournot competition game with demand $p = 100 - q$ and costs for each firm given by $c_i(q_i) = 10q_i$. Imagine that before the two firms play the Cournot game firm 1 can invest in cost reduction. If it invests the costs of firm 1 will drop to $c_1(q_1) = 5q_1$. The cost of investment is $F > 0$. Firm 2 does not have this investment opportunity.
>
> a. Find the value F^* for which the unique subgame-perfect equilibrium involves firm 1 investing.
>
> b. Assume that $F > F^*$. Find a Nash equilibrium of the game that is not subgame perfect.

(a) What is asked. Firm 1 may pay F to cut its unit cost from 10 to 5 before a Cournot duopoly with $p = 100 - q$: (a) the threshold F^* below which the unique SPE has firm 1 investing; (b) for $F > F^*$, a Nash equilibrium that is not subgame perfect.

(b) Method. A two-stage game: an observed investment decision, then a simultaneous Cournot game (Gibbons §2.2.A, pp. 71–73; Tadelis §8.2). SPE: solve the Cournot subgame after each investment decision, then compare at the root (Lecture 4, slide 17; Tadelis Def. 8.3, p. 157). Cournot with asymmetric linear costs: first-order conditions, as in Lecture 2 and Gibbons §1.2.A. Non-SPE Nash equilibria by off-path threats (Lecture 4, slides 8 and 13).

(c) Solution. Notation. Firm 1 chooses I (invest) or N; both firms observe it, then choose $q_1, q_2 \geq 0$. Profits $\pi_i = (100 - q_1 - q_2 - c_i)q_i$, minus F for firm 1 after I.

Step 1: Cournot subgame with costs (c_1, c_2). The first-order conditions are

$$100 - 2q_1 - q_2 - c_1 = 0, \qquad 100 - q_1 - 2q_2 - c_2 = 0,$$

so the best replies are $q_1 = \frac{100 - c_1 - q_2}{2}$, $q_2 = \frac{100 - c_2 - q_1}{2}$, and

$$q_1 = \frac{100 - 2c_1 + c_2}{3}, \qquad q_2 = \frac{100 - 2c_2 + c_1}{3}, \qquad \pi_i = q_i^2. \tag{2}$$

(The last equality uses $p - c_i = q_i$ from the first-order condition.) The subgame equilibrium is unique (the best replies have slope $-\frac{1}{2}$ and cross once).

Subgame	q_1	q_2	p	(π_1, π_2) before F
N: $c_1 = c_2 = 10$	30	30	40	$(900,\ 900)$
I: $c_1 = 5$, $c_2 = 10$	$\frac{100}{3}$	$\frac{85}{3}$	$\frac{115}{3}$	$(\frac{10000}{9},\ \frac{7225}{9})$

Check for I: $q_1 = \frac{100-10+10}{3} = \frac{100}{3}$, $q_2 = \frac{100-20+5}{3} = \frac{85}{3}$, $p = 100 - \frac{185}{3} = \frac{115}{3}$, $p - 5 = \frac{100}{3} = q_1$.

(a) The threshold. Replace each subgame by its value. At the root firm 1 compares $\frac{10000}{9} - F$ (invest) with 900 (not):

$$F^* = \frac{10000}{9} - 900 = \frac{10000 - 8100}{9} = \frac{1900}{9} \approx 211.1.$$

For $F < F^*$ the unique SPE is: firm 1 invests; in each subgame both firms play the Cournot quantities of that subgame. For $F > F^*$ the unique SPE has no investment; at $F = F^*$ both choices are optimal and there are two SPE.

Decomposition (deepening). If firm 2 kept $q_2 = 30$, firm 1's best reply with cost 5 would be $\frac{95-30}{2} = 32.5$, profit $32.5^2 = \frac{4225}{4}$: a *direct* gain of $\frac{4225}{4} - 900 = \frac{625}{4} = 156.25$. Firm 2 in fact cuts its output to $\frac{85}{3}$, adding a *strategic* gain of $\frac{10000}{9} - \frac{4225}{4} = \frac{1975}{36} \approx 54.9$. So $F^* = \frac{625}{4} + \frac{1975}{36} = \frac{1900}{9}$: about a quarter of the value of the investment comes from its effect on the rival.

(b) A Nash equilibrium that is not subgame perfect, for $F > F^*$. A strategy of firm 1 is $(\text{invest?}, q_1^N, q_1^I)$; of firm 2, (q_2^N, q_2^I). Consider

$$\text{firm 1: } (I,\ q_1^N = 0,\ q_1^I = \tfrac{100}{3}), \qquad \text{firm 2: } (q_2^N = 90,\ q_2^I = \tfrac{85}{3}).$$

Firm 2 threatens to flood the market if firm 1 does not invest.

- *Firm 2:* on the path (I) it plays its best reply $\frac{85}{3}$ to $\frac{100}{3}$; its plan after N is never reached, so changing it does not change its payoff.
- *Firm 1:* on the path it earns $\frac{10000}{9} - F$. If it deviates to N, it faces $q_2 = 90$, and its best reply earns $\max_{q_1}(100 - 90 - q_1 - 10)q_1 = \max_{q_1}(-q_1^2) = 0$. So the profile is a Nash equilibrium whenever $\frac{10000}{9} - F \geq 0$.

For $F \in (F^*, \frac{10000}{9}]$ this is a Nash equilibrium in which firm 1 invests although the SPE has no investment. It is not subgame perfect: in the subgame after N, $q_2 = 90$ is not a best reply (against $q_1 = 0$ firm 2's best reply is 45), so the flood is not credible.

For *any* $F > F^*$ (including $F > \frac{10000}{9}$) another example works: firm 1 plays $(N,\ 30,\ 0)$, firm 2 plays $(30,\ 90)$, i.e. it threatens to flood if firm 1 invests. On the path both play Cournot $(30, 30)$; deviating to I would earn at most $\max_{q_1}(100 - 90 - q_1 - 5)q_1 - F = \frac{25}{4} - F < 900$. It is Nash, but the subgame after I is played at $(0, 90)$, which is not a Nash equilibrium of that subgame.

(d) Interpretation. The investment is a sunk, observable commitment, the Brexit lesson of Lecture 4 (slide 11): it pays partly because it changes the rival's behavior (firm 2 retreats from 30 to $\frac{85}{3}$), not only because it lowers firm 1's cost. Because firm 2 cannot credibly promise to respond differently, firm 1's decision depends only on the Cournot continuation values; the non-SPE equilibria show how a non-credible threat by the *rival* could otherwise push firm 1 into, or away from, investment.

(e) Verification. `check_B_L04.py` solves both Cournot subgames with sympy (quantities, profits 900 and $\frac{10000}{9}, \frac{7225}{9}$), computes $F^* = \frac{1900}{9}$ and its decomposition $\frac{625}{4} + \frac{1975}{36}$, and for $F = \frac{1000}{3} \in (F^*, \frac{10000}{9}]$ checks the threat profile: firm 1's best payoff after N against $q_2 = 90$ is 0 (grid of step $\frac{1}{4}$), investing is a best reply, firm 2's on-path output is its best reply, and its best reply after N would be 45, not 90; for the second example, firm 1's best payoff after investing against $q_2 = 90$ is $\frac{25}{4}$ before F. Output line: `tadelis-8-6: F* = 1900/9; flood threat supports investing for F in (F*, 10000/9]`. Tadelis prints no answer.

(f) Common mistakes.

- Computing F^* with firm 2's output held at 30 ($F^* = 156.25$): firm 2 re-optimizes in the new subgame.
- Writing strategies without the off-path quantities: firm 2's strategy must say what it produces after I *and* after N, or no threat can even be stated.
- Giving as a "non-SPE Nash equilibrium" a profile in which a firm is not best-replying on the path (e.g. firm 1 investing when $F > \frac{10000}{9}$).

Answer. (a) $F^* = \frac{10000}{9} - 900 = \frac{1900}{9} \approx 211.1$: for $F < F^*$ the unique SPE has firm 1 investing, then $(q_1, q_2) = (\frac{100}{3}, \frac{85}{3})$; without investment the Cournot play is $(30, 30)$. (b) For $F \in (F^*, \frac{10000}{9}]$: firm 1 invests and plays $\frac{100}{3}$ (and 0 after N); firm 2 plays $\frac{85}{3}$ after I and threatens $q_2 = 90$ after N. Nash, not subgame perfect. (For any $F > F^*$: no investment, Cournot $(30, 30)$, with firm 2 threatening 90 after investment.)

Item 7: Tadelis Exercise 8.4

Exam · difficulty 4/5

Statement (verbatim from the list, item 7)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.4 "Industry Leader", p. 170 [PDF 189]. Reproduced verbatim. (Gibbons Problem 2.6, p. 133, poses the same game and asks only for the subgame-perfect outcome.) Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Industry Leader:** Three oligopolists operate in a market with inverse demand given by $P(Q) = a - Q$, where $Q = q_1 + q_2 + q_3$, and q_i is the quantity produced by firm i. Each firm has a constant marginal cost of production, c, and no fixed cost. The firms choose their quantities dynamically as follows: Firm 1, which is the industry leader, chooses $q_1 \geq 0$. Firms 2 and 3 observe q_1 and then simultaneously choose q_2 and q_3, respectively.
>
> a. How many proper subgames does this dynamic game have? Explain briefly.
>
> b. Is it a game of perfect or imperfect information? Explain briefly.
>
> c. What is the subgame-perfect equilibrium of this game? Show that it is unique.
>
> d. Find a Nash equilibrium that is not a subgame-perfect equilibrium.

(a) What is asked. Leader firm 1 chooses q_1; firms 2 and 3 observe it and choose q_2, q_3 simultaneously; $P = a - Q$, marginal cost c. (a) Number of proper subgames; (b) perfect or imperfect information; (c) the SPE and its uniqueness; (d) a Nash equilibrium that is not subgame perfect.

(b) Method. Proper subgames and perfect information (Lecture 4, slides 14–16; Lecture 3, slide 24; Tadelis Defs. 7.3 and 8.2, pp. 136 and 154). Two-stage game with a simultaneous second stage (Gibbons §2.2.A, pp. 71–73): for each q_1 solve the followers' Cournot game, then let firm 1 optimize (Stackelberg logic, Lecture 3, slide 37; Tadelis §8.3.2, pp. 160–163). Non-SPE Nash equilibria by off-path threats (Lecture 4, slides 8 and 13).

(c) Solution. Notation. $a > c \geq 0$; $q_i \geq 0$; profit $\pi_i = (a - Q - c)\, q_i$ with $Q = q_1 + q_2 + q_3$. A strategy of follower $j \in \{2, 3\}$ is a function $q_j(\cdot)$ of the observed q_1.

(a) Proper subgames. After each $q_1 \geq 0$ the game reaches a node at which firm 2 moves; that node is a singleton information set (everyone knows q_1), and the simultaneous choice of firm 3 lies in an information set contained in what follows it. So every q_1 starts a proper subgame: there is a **continuum** of them (one for each $q_1 \in [0, \infty)$), plus the whole game. No other node starts one: firm 3's nodes belong to an information set that pools all of firm 2's quantities.

(b) Imperfect information. Firms 2 and 3 move simultaneously, so one of them (say firm 3) does not observe the other's choice: its information set contains many nodes. The game has **imperfect information** (complete information, but not perfect).

(c) SPE and uniqueness.

Step 1: the subgame after q_1. If $q_1 < a - c$, the followers' first-order conditions are

$$\frac{\partial \pi_2}{\partial q_2} = a - c - q_1 - 2q_2 - q_3 = 0, \tag{3}$$

$$\frac{\partial \pi_3}{\partial q_3} = a - c - q_1 - q_2 - 2q_3 = 0. \tag{4}$$

Subtracting (4) from (3) gives $q_3 = q_2$; substituting, $a - c - q_1 - 3q_2 = 0$, so

$$q_2(q_1) = q_3(q_1) = \frac{a - c - q_1}{3}.$$

Uniqueness in the subgame. Each profit is strictly concave in the firm's own output, so best replies are $q_j = \max\{0, \frac{a-c-q_1-q_k}{2}\}$. The system (3)–(4) has determinant $2 \cdot 2 - 1 = 3 \neq 0$, so the interior solution is unique; a corner with $q_2 = 0 < q_3$ would need $q_3 = \frac{a-c-q_1}{2}$ and $0 \geq \frac{a-c-q_1-q_3}{2} = \frac{a-c-q_1}{4} > 0$, impossible. The best-reply map has slope $-\frac{1}{2}$, a contraction, so there is no other (also no mixed) equilibrium: the subgame Cournot game is dominance solvable. If $q_1 \geq a - c$, both followers' unique best reply is 0.

Step 2: the leader. For $q_1 < a - c$,

$$\pi_1 = \Big(a - q_1 - 2 \cdot \frac{a - c - q_1}{3} - c\Big) q_1 = \frac{a - c - q_1}{3}\, q_1,$$

strictly concave, with

$$\frac{d\pi_1}{dq_1} = \frac{a - c - 2q_1}{3} = 0 \;\Rightarrow\; q_1^* = \frac{a - c}{2}.$$

For $q_1 \geq a - c$ the price is at most c, so $\pi_1 \leq 0$; the interior choice is the unique global maximum.

Step 3: the SPE.

$$q_1^* = \frac{a - c}{2}, \qquad q_2^*(q_1) = q_3^*(q_1) = \max\Big\{0,\ \frac{a - c - q_1}{3}\Big\}.$$

On the path $q_2 = q_3 = \frac{a-c}{6}$, $Q^* = \frac{5(a-c)}{6}$, $P^* = \frac{a+5c}{6}$, $\pi_1^* = \frac{(a-c)^2}{12}$, $\pi_2^* = \pi_3^* = \frac{(a-c)^2}{36}$. *Uniqueness:* every subgame has a unique Nash equilibrium (Step 1), so in any SPE the followers must use exactly these functions; given them, firm 1's problem has a unique solution (Step 2).

(d) A Nash equilibrium that is not subgame perfect. Let $\hat{q} = \frac{a-c}{4}$ and consider

$$q_1 = \hat{q}, \qquad q_2(q_1) = q_3(q_1) = \begin{cases} \hat{q}, & q_1 = \hat{q}, \\ a - c, & q_1 \neq \hat{q}. \end{cases}$$

- *Followers:* on the path $q_1 = \hat{q}$, each best reply to $(\hat{q}, \hat{q})$ is $\frac{a-c-2\hat{q}}{2} = \frac{a-c}{4} = \hat{q}$. Their plans after other q_1 are off the path and do not affect their payoffs.
- *Leader:* on the path the outcome is three-firm Cournot, $\pi_1 = \frac{(a-c)^2}{16} > 0$. Any $q_1 \neq \hat{q}$ triggers $q_2 = q_3 = a - c$, so $Q \geq 2(a - c)$ and $P - c \leq -(a - c) < 0$: profit ≤ 0. So $\hat{q}$ is a best reply.

It is a Nash equilibrium with outcome $(\frac{a-c}{4}, \frac{a-c}{4}, \frac{a-c}{4})$, not the SPE outcome. It is not subgame perfect: after any $q_1 \neq \hat{q}$ (with $q_1 < a - c$) the pair $(a - c, a - c)$ is not a Nash equilibrium of the subgame (each follower's best reply to $a - c$ is 0).

(d) Interpretation. Moving first and being observed is a commitment: firm 1 produces the monopoly output $\frac{a-c}{2}$ and the followers' joint reply (slope $-\frac{2}{3}$) absorbs part of it, so the leader earns $\frac{(a-c)^2}{12}$ instead of the Cournot $\frac{(a-c)^2}{16}$, and total output rises. The non-SPE equilibrium in (d) shows how the followers could "undo" leadership with a threat to flood the market, but the threat is not credible once q_1 is sunk: this is the Lecture 4 contrast between Nash equilibria held up by threats and the subgame-perfect prediction.

(e) Verification. `check_B_L04.py` solves (3)–(4) and the leader's problem with sympy (determinant 3; $q_1^* = \frac{a-c}{2}$, $q_j^* = \frac{a-c}{6}$, second derivative negative); with $a = 13$, $c = 1$ it iterates the followers' best replies from three starting points for six values of q_1 (including $q_1 \geq a - c$) and

always converges to $\max\{0, \frac{a-c-q_1}{3}\}$; it checks the threat profile of (d): on-path profit 9, every deviation on a grid of step $\frac{1}{4}$ earns ≤ 0, the followers' on-path reply is 3, and flooding is not a best reply. Output line: `tadelis-8-4: SPE q1 = (a-c)/2, q2 = q3 = (a-c)/6; flood-threat NE gives Cournot (a-c)/4 each`. Tadelis prints no answer. The SPE outcome agrees with the 01-04 solution of Gibbons 2.6 (same game).

(f) Common mistakes.

- Counting "two" subgames (one per stage): every value of q_1 starts its own subgame.
- Giving the followers' equilibrium *quantities* $\frac{a-c}{6}$ as their strategies: an SPE needs the reply functions $q_j(q_1)$.
- Solving the followers' problem as a Stackelberg chain (firm 2 before firm 3): here they move simultaneously, so the second stage is a Cournot duopoly on residual demand $a - q_1$.

Answer. (a) A continuum: one proper subgame for each $q_1 \geq 0$, plus the whole game. (b) Imperfect information (firms 2 and 3 move simultaneously). (c) Unique SPE: $q_1^* = \frac{a-c}{2}$, $q_2(q_1) = q_3(q_1) = \max\{0, \frac{a-c-q_1}{3}\}$; outcome $(\frac{a-c}{2}, \frac{a-c}{6}, \frac{a-c}{6})$, $P = \frac{a+5c}{6}$. (d) $q_1 = \frac{a-c}{4}$; followers play $\frac{a-c}{4}$ if $q_1 = \frac{a-c}{4}$ and $a - c$ otherwise: Nash, not subgame perfect.

Item 8: Muñoz-Garcia Ch. 4, Exercise 12

Exam · difficulty 3/5

Statement (verbatim from the list, item 8)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 4, Exercise 12 "Stackelberg Competition with 2, or 3 Firms", pp. 185–186 [PDF 197–198]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider a Stackelberg model of output competition where firm 1 is the industry leader and firm 2 is the follower. Firms face a linear inverse demand curve $p(Q) = 10 - Q$, where $Q = q_1 + q_2$ denotes the aggregate output sold by both firms. In addition, consider that firms are cost symmetric, facing a common marginal cost of $MC_i = 2$ for every firm $i = \{1, 2\}$ and no fixed costs.
>
> (a) Find the equilibrium in this Stackelberg game.
>
> (b) Assume now that a new firm (firm 3) joins the industry, acting last (after firm 2 chooses its output). Find the equilibrium in this Stackelberg game. Compare your results with those in part (a). How is the output of firm 1 and 2 affected by firm 3's entry?

(a) What is asked. Stackelberg with $p(Q) = 10 - Q$ and marginal cost 2 for each firm: (a) the equilibrium with a leader (firm 1) and a follower (firm 2); (b) the equilibrium when firm 3 joins and moves last, and how firm 3's entry changes the outputs of firms 1 and 2.

(b) Method. Backward induction with a continuum of actions, from the last mover to the first (Tadelis §8.3.2, pp. 160–163; SPE, Lecture 4, slides 31–33): each mover's best reply is a function of the outputs already chosen, which the earlier movers substitute into their own problems. Every output is assumed interior; this is checked at the end.

(c) Solution. Write $\mu = 10 - 2 = 8$ for the margin when $Q = 0$, so each firm's profit is $(8 - Q)\, q_i$.

(a) Two firms. *Follower.* Given q_1, firm 2 maximizes $(8 - q_1 - q_2)\, q_2$; the first-order condition $8 - q_1 - 2q_2 = 0$ gives

$$q_2(q_1) = 4 - \frac{q_1}{2}.$$

Leader. Firm 1 maximizes $(8 - q_1 - q_2(q_1))q_1 = (4 - \frac{q_1}{2})q_1$, with first-order condition $4 - q_1 = 0$, so $q_1^* = 4$ and $q_2^* = 4 - 2 = 2$. Then $Q = 6$, $p = 4$, and the profits are $\pi_1 = (4 - 2) \cdot 4 = 8$ and $\pi_2 = (4 - 2) \cdot 2 = 4$.

(b) Three firms. Firm 3 moves after observing q_1, q_2; firm 2 after q_1 only.

1. *Firm 3.* It maximizes $(8 - q_1 - q_2 - q_3)\, q_3$, so

$$q_3(q_1, q_2) = \frac{8 - q_1 - q_2}{2} = 4 - \frac{q_1 + q_2}{2}.$$

2. *Firm 2,* anticipating $q_3(q_1, q_2)$. The margin it faces is $8 - q_1 - q_2 - q_3 = 8 - q_1 - q_2 - 4 + \frac{q_1 + q_2}{2} = 4 - \frac{q_1 + q_2}{2}$, so it maximizes $(4 - \frac{q_1}{2} - \frac{q_2}{2})q_2$, with first-order condition $4 - \frac{q_1}{2} - q_2 = 0$:

$$q_2(q_1) = 4 - \frac{q_1}{2}, \qquad q_3(q_1) = 4 - \frac{q_1 + 4 - q_1/2}{2} = 2 - \frac{q_1}{4}.$$

Firm 2's reply to the leader is the same as in part (a).

3. *Firm 1,* anticipating both. The margin it faces is $8 - q_1 - \left(4 - \frac{q_1}{2}\right) - \left(2 - \frac{q_1}{4}\right) = 2 - \frac{q_1}{4}$, so it maximizes $\left(2 - \frac{q_1}{4}\right)q_1$, with first-order condition $2 - \frac{q_1}{2} = 0$, so $q_1^* = 4$.

Then $q_2^* = 4 - 2 = 2$ and $q_3^* = 2 - 1 = 1$. Total output is $Q = 7$ and the price $p = 3$; the profits are $\pi_1 = (3-2)\cdot 4 = 4$, $\pi_2 = (3-2)\cdot 2 = 2$, $\pi_3 = (3-2)\cdot 1 = 1$. All outputs are positive, so the interior assumption holds. The SPE strategies are the reply functions $q_3(q_1, q_2)$ and $q_2(q_1)$ together with $q_1^* = 4$, defined for every output of the earlier movers.

Comparison.

	q_1	q_2	q_3	Q	p	(π_1, π_2, π_3)
(a) two firms	4	2	—	6	4	(8, 4, −)
(b) three firms	4	2	1	7	3	(4, 2, 1)

Firm 3's entry changes neither q_1 nor q_2: the earlier movers keep producing 4 and 2. Total output rises from 6 to 7 and the price falls from 4 to 3, so the profits of firms 1 and 2 are halved.

(d) Interpretation. With linear demand and constant cost, a last mover who replies $q_3 = \frac{8 - q_1 - q_2}{2}$ turns every earlier mover's margin into exactly half of what it was: firm 2 faces $\frac{1}{2}(8 - q_1 - q_2)$ instead of $8 - q_1 - q_2$, and firm 1 faces $2 - \frac{q_1}{4}$ instead of $4 - \frac{q_1}{2}$. Multiplying an objective by $\frac{1}{2}$ does not move its maximizer, so q_1^* and q_2^* are unchanged, while the entrant adds output and lowers the price. First-mover status protects an incumbent's output, not its profit.

(e) Verification. `check_L04.py` solves the two-firm and three-firm Stackelberg games by backward induction with sympy (symbolic reply functions, then the leader), confirms $q = (4, 2)$ and $(4, 2, 1)$, the prices 4 and 3 and the profits, and verifies on an integer grid that no firm gains by a unilateral deviation along the SPE reply functions. Output line: `munoz-garcia-4-12: (a) q = (4, 2), p = 4; (b) q = (4, 2, 1), p = 3`. *Comparison with Muñoz-Garcia's answer* (pp. 186–189): it agrees on every figure: part (a) $q_1^* = 4$, $q_2^* = 2$; part (b) $q_1^* = 4$, $q_2^* = 2$, $q_3^* = 1$ and "firms 1 and 2 continue to produce 4 and 2 units of output respectively as in the 2-player game, while firm 3 produces 1 unit". The book does not report prices or profits.

(f) Common mistakes.

- Solving firm 1's problem with firm 2's reply from part (a) in part (b): firm 1 must anticipate both later movers.
- Treating firms 2 and 3 as moving simultaneously (a Cournot game after q_1): the exercise has firm 3 observe q_2.
- Concluding from q_1, q_2 unchanged that entry is harmless: the price and every incumbent's profit fall.

Answer. (a) $q_1^* = 4$, $q_2^* = 2$, $p = 4$, profits $(8, 4)$. (b) $q_1^* = 4$, $q_2^* = 2$, $q_3^* = 1$, $p = 3$, profits $(4, 2, 1)$; firm 3's entry leaves the outputs of firms 1 and 2 unchanged, raises total output from 6 to 7 and halves their profits.

Item 9: Tadelis Exercise 8.15

Exam · difficulty 3/5

Statement (verbatim from the list, item 9)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.15 "Time Inconsistency", p. 174 [PDF 193]. Reproduced verbatim. *Note (not in the book):* the example of Section 8.3.4 is on pp. 166–169; its printed first-period choice $x_1 = \frac{2}{3}K$ (p. 169) does not follow from the objective printed there (the maximum is at $\frac{K}{2}$); see `Notas/L03/L03-08-time-inconsistency.md` §2.2. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Time Inconsistency:** Consider the three-period example of a player with hyperbolic discounting described in Section 8.3.4 with utility $\ln(x)$ in each of the three periods, with initial budget K and with discount factors $\delta = 1$ and $\beta = \frac{1}{2}$.
>
> a. Solve the optimal plan of action of a naive player 1 who does not take into account how his future self, player 2, will alter the plan. What is player 1's optimal plan x_1^*, x_2^*, and x_3^* as a function of K?
>
> b. Let K_2 be the amount left from the solution to (a) after player 1 consumes his planned choice of x_1^*. Given K_2, what is the optimal plan of player 2? In what way does it differ from the optimal plan set out by player 1?

Referred to in the statement. The three-period example of Section 8.3.4 (Tadelis pp. 166 and 168 [PDF p. 185 and 187]) (reproduced from the book):

> Consider a player with $u(x) = \ln(x)$ who needs to allocate a fixed budget K across three periods. Since he will not waste any of his budget, it follows that $x_1 = K - x_2 - x_3$ [...]
>
> To model this idea consider a different discounting rule known as *hyperbolic discounting*. The player uses the discount rate δ for the future periods as in exponential discounting, but he uses an *additional* discount factor $\beta \in (0, 1)$ to discount all of the future compared to present consumption. Using our three-period example, the player's modified discounted present-value problem is given by
>
> $$\max_{x_2, x_3} v(K - x_2 - x_3, x_2, x_3) = \ln(K - x_2 - x_3) + \beta\delta \ln(x_2) + \beta\delta^2 \ln(x_3). \tag{8.9}$$
>
> Simply put, when the player is looking toward the future, the discount factor he uses between periods $t = 1$ and $t = 2$ is stronger than the one he uses between periods $t = 2$ and $t = 3$. In particular, at any period and looking forward, the player uses $\beta\delta < \delta$ to discount between the current period and the next, while he uses only δ to discount between any two consecutive future periods.[4] As we will now see, hyperbolic discounting will cause problems of self-control, in that a player will plan to do one thing but later choose to revise his plan.
>
> Imagine that the player is very well aware of his self-control problem—at period $t = 1$ he discounts between periods $t = 2$ and $t = 3$ at the rate of δ, but he knows that when tomorrow ($t = 2$) comes around, he will discount $t = 3$ at a rate of $\beta\delta$. In other words, the player at $t = 1$ (call him player 1) is aware that he is playing a game with his "future self" at $t = 2$ (call him player 2), who will exhibit different preferences over time. In this case a rational, forward-looking player 1 will solve his problem using backward induction; in other words he will solve for the subgame-perfect equilibrium of the implied dynamic game.
>
> 4. More generally the discounted present value of a player who uses hyperbolic discounting is given by $u(x_1) + \beta \sum_{t=2}^{T} \delta^{t-1} u(x_t)$.

(a) What is asked. Three periods, utility $\ln x$ in each, budget K, $\delta = 1$, $\beta = \frac{1}{2}$. (a) The plan (x_1^*, x_2^*, x_3^*) of a *naive* player 1, who ignores that her future self will change it; (b) with K_2 the budget left after x_1^*, the plan player 2 actually chooses, and how it differs.

(b) Method. Present-biased (quasi-hyperbolic) discounting, Tadelis §8.3.4, pp. 166–169: in period t the self maximizes $u(x_t) + \beta \sum_{s>t} \delta^{s-t} u(x_s)$, so the weights of periods $2, 3$ seen from period 1 are $\beta\delta = \frac{1}{2}$ and $\beta\delta^2 = \frac{1}{2}$, and seen from period 2 the weight of period 3 is $\beta\delta = \frac{1}{2}$. A naive player optimizes today's objective over the whole plan (a one-person decision problem); then the next self re-optimizes, which is the single-agent version of "a plan that is not credible" (Lecture 4, slides 22–23, applied to one's own future choices). This is reading, not in the slides of Lecture 4.

(c) Solution. (a) Player 1's plan. She maximizes $\ln x_1 + \frac{1}{2}\ln x_2 + \frac{1}{2}\ln x_3$ subject to $x_1 + x_2 + x_3 = K$. With the budget substituted, $x_1 = K - x_2 - x_3$, the first-order conditions are

$$-\frac{1}{K - x_2 - x_3} + \frac{1/2}{x_2} = 0, \qquad -\frac{1}{K - x_2 - x_3} + \frac{1/2}{x_3} = 0,$$

so $x_2 = x_3 = \frac{1}{2}x_1$. The budget gives $x_1 + \frac{1}{2}x_1 + \frac{1}{2}x_1 = K$, hence

$$x_1^* = \frac{K}{2}, \qquad x_2^* = \frac{K}{4}, \qquad x_3^* = \frac{K}{4}.$$

(The objective is strictly concave, so this stationary point is the maximum.)

(b) Player 2's plan. After consuming x_1^*, the budget is $K_2 = K - x_1^* = \frac{K}{2}$. Player 2 maximizes $\ln x_2 + \frac{1}{2}\ln x_3$ with $x_3 = K_2 - x_2$: the first-order condition $\frac{1}{x_2} - \frac{1/2}{K_2 - x_2} = 0$ gives $K_2 - x_2 = \frac{1}{2}x_2$, so

$$x_2 = \frac{2}{3}K_2 = \frac{K}{3}, \qquad x_3 = \frac{1}{3}K_2 = \frac{K}{6}.$$

Difference. Player 1 planned $(\frac{K}{4}, \frac{K}{4})$, an equal split of what remains, because from her viewpoint periods 2 and 3 are equally discounted ($\delta = 1$). Player 2 consumes more now, $\frac{K}{3} > \frac{K}{4}$, and leaves $\frac{K}{6} < \frac{K}{4}$ for period 3, because for her period 3 is the discounted future ($\beta\delta = \frac{1}{2}$) and period 2 is the present. The plan is time-inconsistent: player 1's plan is not carried out.

Benchmark. With exponential discounting ($\beta = 1$, $\delta = 1$) the plan is $K/3$ in every period, and the period-2 self would carry it out.

(d) Interpretation. The two selves disagree about the split of the future. The disagreement arises from $\beta < 1$ alone: it makes the *current* self weigh today against all later dates, and later selves repeat the same bias when they become the current self. A naive planner is surprised again and again; a sophisticated one (Tadelis 8.16) foresees the revision and may want to commit.

(e) Verification. `check_L04.py` maximizes the three objectives with sympy (the $\delta = 1$, $\beta = \frac{1}{2}$ case and the exponential benchmark), confirms $(\frac{K}{2}, \frac{K}{4}, \frac{K}{4})$, then $K_2 = \frac{K}{2}$, player 2's plan $(\frac{K}{3}, \frac{K}{6})$, and checks by a grid search at $K = 90$ that these are the maximizers. Output line: `tadelis-8-15: naive plan (K/2, K/4, K/4); self-2 plan (K/3, K/6)`. Tadelis prints no answer.

(f) Common mistakes.

- Solving the *sophisticated* problem (substituting player 2's reply into player 1's objective, which gives $(\frac{K}{2}, \frac{K}{3}, \frac{K}{6})$): part (a) asks for the naive plan, in which player 1 ignores the revision.

- Weighting period 3 by $\beta\delta^2$ in player 2's problem: from period 2 the weight is $\beta\delta$.
- Using $\beta = \frac{1}{2}$ for the first term as well: today's utility has weight 1.

Answer. (a) $x_1^* = \frac{K}{2}$, $x_2^* = x_3^* = \frac{K}{4}$. (b) $K_2 = \frac{K}{2}$; player 2 chooses $x_2 = \frac{K}{3}$, $x_3 = \frac{K}{6}$: she consumes more in period 2 and leaves less for period 3 than player 1 planned.

Item 10: Tadelis Exercise 8.7

Advanced · difficulty 4/5

Statement (verbatim from the list, item 10)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.7 "Debt and Repayment", p. 171 [PDF 190]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Debt and Repayment:** A project costing \$100 yields a gross return of \$110. A lender (player 1) is approached by a debtor (player 2) requesting a standard loan contract to complete the project. If the lender chooses *not to offer* a loan, then both parties earn nothing. If the lender chooses to *offer* a loan of \$100, the debtor can realize the project's gains and is obliged by contract to repay \$105. For simplicity assume that money is continuous and that the debtor can choose to return any amount of money $x \leq 110$. Ignore the time value of money. Assume first that no legal system is in place that can compel the debtor to repay, so that default on the loan (less than full repayment) carries no repercussions for the debtor.
>
> a. Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?
>
> b. Now assume that there is a legal system in place that allows the lender to *voluntarily* choose whether or not to sue when the debtor defaults and repays an amount $x < 105$. Furthermore assume that it is *costless* to use the legal system (it is supplied by the state), and if the lender sues a debtor that defaulted, the lender will get the \$105 repaid in full. After paying the lender, the borrower will pay a fine of \$5 to the court, above and beyond the repayment. Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?
>
> c. Are there Nash equilibria in the game described in (b) that are not subgame-perfect equilibria?
>
> d. Now assume that using the legal system is *costly:* if the lender sues, he pays lawyers a *legal fee* of \$105 (this is the lawyers' price; it is unrelated to the contract). The rest of the game proceeds as before. (If the lender sues a debtor that defaulted, the lender will get repaid in full; after paying the lender, the borrower will pay a fine of \$5 above and beyond the repayment.) Model this as an extensive-form game tree as best you can and find a subgame-perfect equilibrium of this game. Is it unique?
>
> e. Are there Nash equilibria in the game described in (d) that are not subgame-perfect equilibria?
>
> f. Now assume that a change in the law is proposed: upon default, if a debtor is sued he has to first repay the lender \$105 and then pay the legal fees of \$105 above and beyond repayment of the loan, and no extra fine is imposed. Should the lender be willing to pay to have this change in the law enacted? If so, how much?
>
> g. If you were a social planner, would you implement the suggested law?

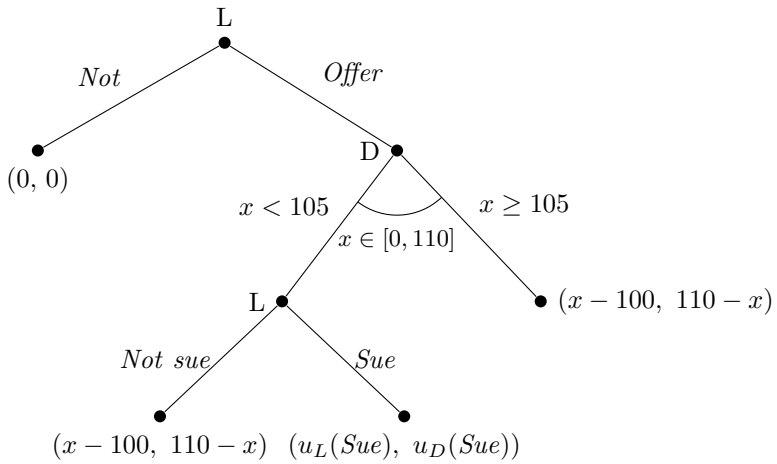
(a) What is asked. For the loan game with no courts, free courts and costly courts: the trees and SPE, with uniqueness ((a), (b), (d)); Nash equilibria that are not SPE ((c), (e)); what the lender would pay for a "loser pays the legal fees" law (f); and whether a planner should adopt it (g).

(b) Method. Backward induction with a continuum of actions: for every repayment x, solve the lender's suit decision; then the debtor's choice of x; then the lender's offer (Tadelis §8.1, Def. 8.1, p. 152; SPE Def. 8.3, p. 157; the Fact on p. 158). Nash equilibria that are not SPE rely on plans

at nodes off the equilibrium path (Def. 7.8, p. 146; §8.2, pp. 157–158). Lecture 4: credible and non-credible threats, slides 8, 11 and 13; SPE, slide 17.

(c) Solution. Set-up and reading of the statement. Player 1 (lender, L) chooses *Offer* or *Not*. After *Offer*, player 2 (debtor, D) repays $x \in [0, 110]$. Repaying a negative amount is not an option, so $x \geq 0$. Without a suit, $(u_L, u_D) = (x - 100,\ 110 - x)$. When courts exist and $x < 105$, the lender then chooses *Sue* or *Not sue*. A suit brings the total repayment to 105, with the x already paid counting toward it. The debtor then pays the fine or the fee on top. The three legal regimes differ only in the payoffs after *Sue*:

	(b) free courts, fine 5	(d) lender pays fee 105	(f) debtor pays fee 105
$u_L(Sue)$	$105 - 100 = 5$	$105 - 100 - 105 = -100$	$105 - 100 = 5$
$u_D(Sue)$	$110 - 105 - 5 = 0$	$110 - 105 - 5 = 0$	$110 - 105 - 105 = -100$



Payoffs (u_L, u_D). The arc marks the continuum of repayments x. The lender's node after $x < 105$ exists only in (b), (d) and (f); in (a) every x ends the game with $(x - 100, 110 - x)$.

A pure strategy of the lender is an offer decision plus a suit rule $sue(x)$ for every $x < 105$: a function, not a single choice. A pure strategy of the debtor is a number x.

(a) No legal system. *Debtor:* $110 - x$ is strictly decreasing, so $x^* = 0$, giving $(u_L, u_D) = (-100, 110)$. *Lender: Not* gives $0 > -100$. SPE: $(Not;\ x = 0)$, payoffs $(0, 0)$. It is unique, since both maximizers are unique.

(b) Costless courts. *Lender after a default $x < 105$: Sue* gives $5 > x - 100$, so he sues after every default. *Debtor:* any $x < 105$ is followed by a suit and gives 0; $x \in [105, 110]$ gives $110 - x \leq 5$, with the maximum at $x = 105$. Since $5 > 0$, $x^* = 105$. *Lender at the root: Offer* gives $105 - 100 = 5 > 0$. SPE:

$$\big(Offer,\ sue(x) = Sue\ \forall x < 105;\ \ x = 105\big), \qquad (u_L, u_D) = (5, 5).$$

It is unique: every comparison is strict and $x = 105$ is the only maximizer.

(c) Nash equilibria of (b) that are not SPE: none in pure strategies.

- *No pure NE without lending.* Whatever x the debtor plans, the lender can switch to *Offer* and to a suit rule that is best at that x. He then gets 5 if $x < 105$ (sue) or $x - 100 \geq 5$ if $x \geq 105$, more than 0. So *Not* is never a best reply. Unlike a threat against the opponent, a lender's plan "I would not sue" cannot keep him out: he controls it.

- *With lending, the profile is forced.* If $x > 105$ the debtor gains by moving to 105. If $x < 105$ the lender gains by suing (if he does not) or the debtor gains by moving to 105 (if he does: $5 > 0$). So $x = 105$. If the suit rule skipped even one $x' < 105$, the debtor would move there and get $110 - x' > 5$. So the rule must be "sue after every $x' < 105$", which is the SPE.

So in pure strategies the only Nash equilibrium is the SPE. With behavioural strategies the answer changes, though not the outcome. The lender may randomize at unreached defaults and sue after x' with probability $s(x')$, provided the debtor still prefers 105: $(1 - s(x'))(110 - x') \leq 5$, i.e. $s(x') \geq 1 - \frac{5}{110-x'}$ (e.g. $\frac{1}{6}$ at $x' = 104$). Any $s(x') < 1$ is not sequentially rational, since suing is strictly better. These are Nash equilibria that are not SPE, with the same path and payoffs $(5, 5)$.

(d) Costly courts (lender pays the fee). *Lender after a default $x < 105$: Sue* gives -100 and *Not sue* gives $x - 100$. Not suing is strictly better for every $x > 0$, and at $x = 0$ the two tie ($-100 = -100$). *Debtor:* if the lender does not sue at $x = 0$, the debtor's payoff $110 - x$ on $[0, 105)$ is maximized at $x = 0$, giving 110, which beats $110 - x \leq 5$ on $[105, 110]$. Could the lender instead sue at $x = 0$, with probability $s > 0$? The debtor would get $110(1 - s) < 110$ at $x = 0$ and $110 - x$ at every small $x > 0$. The supremum 110 is then not attained, so the debtor has no best reply. The tie at 0 must therefore be broken toward *Not sue*. *Lender at the root: Offer* gives $-100 < 0$. The unique SPE is

$$(\textit{Not},\ \textit{sue}(x) = \textit{Not sue}\ \forall x < 105;\quad x = 0), \qquad (u_L, u_D) = (0, 0),$$

the outcome of (a). A legal system that is too expensive to use is the same as none.

(e) Nash equilibria of (d) that are not SPE: yes.

- *Lending supported by a non-credible threat:*

  $$(\textit{Offer},\ \textit{sue}(x) = \textit{Sue}\ \forall x < 105;\quad x = 105), \qquad (u_L, u_D) = (5, 5).$$

  It is a NE. The debtor gets 5 at $x = 105$ and 0 at any default, since he is sued. The lender gets $5 > 0$, and no suit node is reached, so the suit rule costs him nothing. It is not an SPE. In the subgame after any default $x > 0$, suing gives $-100 < x - 100$. As in (c), this is the only NE with lending in pure strategies: one gap in the suit rule invites the debtor to default there. It is better for *both* players than the SPE $(0, 0)$. The threat is what would make lending possible, but it is not credible.

- *No lending, with other plans: Not*, any planned $x \leq 100$ and any suit decision at that x. Offering would give the lender at most $\max\{-100,\ x-100\} = x-100 \leq 0$. All of these except the SPE are Nash equilibria that are not subgame perfect, e.g. $(\textit{Not},\ \text{never sue};\ x = 50)$, because $x = 50$ is not the debtor's best reply after *Offer*.

(f) Loser pays the fees. After a default $x < 105$, *Sue* gives the lender $105 - 100 = 5 > x - 100$: the suit is now credible. The debtor gets -100 from any default and $110 - x$ from $x \in [105, 110]$, so $x = 105$ ($5 > -100$). Offering gives $5 > 0$. Unique SPE: $(\textit{Offer},\ \textit{Sue}\ \forall x < 105;\ x = 105)$, payoffs $(5, 5)$. Compared with (d) the lender's equilibrium payoff rises from 0 to 5. Anything he pays for the reform is sunk before the game starts and does not change the SPE, so he gains iff the price is below \$5. He would pay *up to \$5*.

(g) The planner. Total surplus is 0 without the law (no loan) and $110 - 100 = 10$ with it. On the equilibrium path nobody defaults, so nobody is sued and the \$105 fee is never paid. The reform moves both players from $(0, 0)$ to $(5, 5)$, a Pareto improvement, so the planner should adopt it if enacting it costs less than \$10 (the debtor too would pay up to \$5). The harsh off-path penalty (-100 for the debtor) has no social cost in this model because it is never imposed. It would have one if default could happen by bad luck, which the model rules out.

(d) Interpretation. The project creates \$10 of value, but the lender can capture it only if the debtor believes default will be punished. In (a) and (d) the debtor's promise to repay is not

credible, and in (d) neither is the lender's threat to sue, because a suit costs more than it recovers. Lending collapses even though both would gain. The Nash equilibrium in (e) shows the gains a credible threat would deliver. The SPE shows why the threat by itself does not work. The reform in (f) does not change what happens on the path; it changes who pays for enforcement *off* the path. That makes suing the lender's best reply to default, so the threat becomes credible and is never used. This is the Brexit lesson in a credit market (slides 11 and 13): the institution that makes a threat credible creates value, while announcing the threat creates none. Free courts in (b) and loser-pays in (f) give the same outcome.

(e) Verification. `check_D.py` encodes the payoffs of the four regimes exactly and works on the repayment grid $x \in \{k/8\}$, which contains 0, 100, 104.5 and 105. It computes the sequentially rational suit choice at every default, showing the tie at $x = 0$ in (d) and strict preferences elsewhere. It then computes the debtor's best x and the lender's offer. It tests the claimed Nash equilibria with unilateral deviations: the lender's joint offer/suit deviations and every debtor x. It confirms that a single gap in the suit rule breaks the threat equilibria and that no no-lending NE exists in (b). It also checks the behavioural bound $s(x') \geq 1 - \frac{5}{110-x'}$. Output: "Tadelis 8.7: (a) no loan, x = 0, (0,0); (b) loan, x = 105, sue every x < 105, (5,5); (c) pure NE = SPE; (d) no loan, x = 0, never sue (tie at x = 0), (0,0); (e) NE with loan and threat to sue, (5,5), and no-loan NE with x <= 100; (f) lender pays up to 5; (g) surplus +10". Tadelis prints no answer.

(f) Common mistakes.

- Reading (c) as "the lender threatens not to sue, so he does not lend". The lender's own off-path plan cannot discipline the lender: he can change his whole strategy, offer and suit rule together. In (b) every Nash equilibrium in pure strategies lends.
- In (d), leaving the tie at $x = 0$ unresolved, or claiming an SPE where the lender sues only at $x = 0$. Then the debtor has no best reply, so that profile cannot be part of an equilibrium.
- In (f) and (g), counting the \$105 fee as a cost of the reform. It is paid only after a default, which never happens in equilibrium.

Answer. (a) SPE (*Not*; $x = 0$), payoffs $(0, 0)$, unique. (b) SPE: offer, repay $x = 105$, sue after every $x < 105$; payoffs $(5, 5)$, unique. (c) None in pure strategies: the only pure Nash equilibrium is the SPE (behavioural NE with off-path suit probability $s(x') \in [1 - \frac{5}{110-x'}, 1)$ exist, with the same outcome). (d) SPE: no loan, $x = 0$, never sue; payoffs $(0, 0)$, unique (the tie at $x = 0$ must go to not suing). (e) Yes: (offer, sue after every default; $x = 105$), payoffs $(5, 5)$, rests on a non-credible threat to sue; so do the other no-loan profiles with planned $x \leq 100$. (f) Yes, up to \$5 (his SPE payoff goes from 0 to 5). (g) Yes: surplus rises from 0 to 10, both gain \$5, and the fee is never paid in equilibrium.

Item 11: Tadelis Exercise 8.12

Advanced · difficulty 4/5

Statement (verbatim from the list, item 11)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.12 "Agenda Setting", p. 173 [PDF 192]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Agenda Setting:** An agenda-setting game is described as follows. The "issue space" (set of possible policies) is an interval $X = [0, 5]$. An agenda setter (player 1) proposes an alternative $x \in X$ against the status quo $q = 4$. After player 1 proposes x, the legislator (player 2) observes the proposal and selects between the proposal x and the status quo q. Player 1's most-preferred policy is 1, and for any final policy $y \in X$ his payoff is given by
>
> $$v_1(y) = 10 - |y - 1|,$$
>
> where $|y - 1|$ denotes the absolute value of $(y - 1)$. Player 2's most preferred policy is 3, and for any final policy $y \in X$ his payoff is given by
>
> $$v_2(y) = 10 - |y - 3|.$$
>
> That is, each player prefers policies that are closer to his most-preferred policy.
>
> a. Write the game down as a normal-form game. Is this a game of perfect or imperfect information?
>
> b. Find a subgame-perfect equilibrium for this game. Is it unique?
>
> c. Find a Nash equilibrium that is not subgame perfect. Is it unique? If yes, explain. If not, show all the Nash equilibria for this game.

(a) What is asked. An agenda setter (player 1) proposes $x \in X = [0, 5]$ against the status quo $q = 4$; the legislator (player 2) picks x or q. Payoffs $v_1(y) = 10 - |y - 1|$, $v_2(y) = 10 - |y - 3|$ for the final policy y. (a) The normal form and whether the game has perfect information; (b) a subgame-perfect equilibrium and its uniqueness; (c) a Nash equilibrium that is not subgame perfect, and all Nash equilibria.

(b) Method. Backward induction with a continuum of proposals (Tadelis §8.1–8.2, pp. 151–159; SPE, Lecture 4, slides 31–33): player 2's best reply to every proposal, then player 1's best proposal; the Nash equilibria that are not SPE use off-path rejections as non-credible threats (Lecture 4, slides 22–23).

(c) Solution. **(a) Strategies and information.** Player 1's strategy is a proposal $x \in [0, 5]$. Player 2 observes it, so a strategy of player 2 is a map $x \mapsto \{\text{accept}, \text{reject}\}$, equivalently the set $A \subseteq X$ of proposals he accepts: S_2 is the set of subsets of X. If the final policy is $y(x, A) = x$ if $x \in A$ and $y = 4$ otherwise, the payoffs are $v_i(y(x, A))$. Player 2 observes player 1's move, and each player moves once at a singleton information set, so this is a game of *perfect* information, with a continuum of subgames, one for each proposal.

(b) The SPE. *Player 2.* He accepts x iff $v_2(x) \geq v_2(4)$, i.e. $|x - 3| \leq |4 - 3| = 1$, that is $x \in [2, 4]$; he is indifferent at $x = 2$ and $x = 4$ and strictly rejects every other proposal. At $x = 4$ accepting and rejecting give the same policy, 4. *Player 1.* Accepted proposals give him $v_1(x) = 11 - x$ for

$x \in [2,4]$, decreasing in x; a rejected proposal gives $v_1(4) = 7$. The best is the smallest acceptable proposal, $x = 2$, with $v_1 = 9$, provided player 2 accepts at the indifference point $x = 2$. If he rejected at $x = 2$, player 1's problem would have no solution (the infimum 2 of the acceptance set is not attained), so no SPE would exist; hence the SPE requires that player 2 *accepts* at $x = 2$.

So the SPE outcome is unique: proposal $x^* = 2$, accepted, final policy $y = 2$, payoffs $(v_1, v_2) = (9, 9)$. The SPE strategy of player 2 is: accept every $x \in [2,4]$ except possibly the irrelevant tie at $x = 4$ (it leads to the same policy), reject all others. Player 2's strategy is therefore unique up to his response to the proposal $x = 4$; the outcome and the payoffs are unique.

(c) Nash equilibria. A profile (x^*, A) is a Nash equilibrium iff (i) player 2's reply is optimal at the *proposal actually made* and (ii) player 1 cannot gain by proposing something else, given A. Off the path, player 2's replies are unconstrained, so he may reject proposals that would be good for player 1.

- *Outcome $y \in [2,4]$ through acceptance.* For any $y \in [2,4]$, take $x^* = y$ and $A = \{y\}$: player 2 accepts y, which is optimal because $|y - 3| \leq 1$, and every other proposal is rejected and yields the status quo, $v_1(4) = 7$. Player 1's payoff from $x^* = y$ is $11 - y \geq 7$, so he does not deviate. Each such pair is a Nash equilibrium. None is subgame perfect: after a proposal $x \in (2,4)$ player 2 strictly prefers accepting ($v_2(x) > v_2(4) = 9$), yet $A = \{y\}$ rejects it, a threat he would not carry out. A subgame-perfect A must contain $(2,4)$ and nothing outside $[2,4]$; then player 1 proposes 2, so the only subgame-perfect equilibrium is the one of part (b).
- *Outcome 4 through rejection.* Take $x^* \in [0,2] \cup [4,5]$ (so rejecting is optimal for player 2) and $A \cap [0,4) = \varnothing$: nothing player 1 can propose below 4 is accepted, so the status quo stays, with payoffs $(7,9)$. Player 1 cannot gain, since the only acceptable proposals are in $[4,5]$, which give him at most 7.

Hence there is a continuum of Nash equilibria whose final policies are exactly $y \in [2,4]$ (no Nash equilibrium produces a policy outside $[2,4]$, since an accepted x must satisfy $|x - 3| \leq 1$ and the alternative is 4). Examples that are not subgame perfect: $(x^*, A) = (3, \{3\})$ with final policy 3, payoffs $(8,10)$; and $(x^*, A) = (0, \varnothing)$ with final policy 4, payoffs $(7,9)$.

(d) Interpretation. Without subgame perfection the legislator can hold the setter to any policy in $[2,4]$ by threatening to reject every better offer; credibility removes those threats and leaves the setter his best policy among those the legislator would truly accept, 2, the policy that makes the legislator just indifferent to the status quo (the mirror image of $q = 4$ around his ideal point 3). The setter's power comes from the proposal being a take-it-or-leave-it offer, with the status quo as the legislator's outside option.

(e) Verification. `check_L04.py` works on the grid $X = \{0, \frac{1}{2}, \ldots, 5\}$ and enumerates all 2^{11} acceptance sets and all proposals: the set of final policies of Nash equilibria is exactly the grid points of $[2,4]$ (and no Nash equilibrium gives a policy outside it); player 2's best replies accept exactly when $|x - 3| \leq 1$; the SPE outcome is 2 when the tie at 2 is accepted, and $2 + h$ (no maximum in the continuum) otherwise, for grid steps $h = \frac{1}{2}, \ldots, \frac{1}{64}$. Output line: `tadelis-8-12: SPE outcome x = 2 (payoffs 9, 9); NE outcomes = [2, 4]`. Tadelis prints no answer.

(f) Common mistakes.

- Letting player 2 accept every proposal that is better than q for *player 1*: the legislator compares policies with his own payoff, $|x - 3| \leq 1$.
- Proposing $x = 1$ (player 1's ideal point): player 2 rejects it, since $|1 - 3| = 2 > 1$.

- Claiming a unique Nash equilibrium: off-path rejections generate a continuum of equilibria with final policies in $[2, 4]$.

Answer. (a) Strategies: a proposal x for player 1, an acceptance set $A \subseteq X$ for player 2; perfect information. (b) Unique SPE outcome: $x^* = 2$ accepted, payoffs $(9, 9)$; player 2 accepts exactly the proposals in $[2, 4]$. (c) Not unique: for every $y \in [2, 4]$ there is a Nash equilibrium with final policy y (e.g. $x^* = 3$, $A = \{3\}$) and one with the status quo 4 ($x^* = 0$, $A = \varnothing$); only $y = 2$ is subgame perfect.

Item 12: Tadelis Exercise 8.16

Advanced · difficulty 4/5

Statement (verbatim from the list, item 12)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 8.16 "The Value of Commitment", p. 174 [PDF 193]. Reproduced verbatim. *Note (not in the book):* "We solved the optimal consumption plan" refers to pp. 168–169, whose printed $x_1 = \frac{2}{3}K$ is a slip; with the objective as printed the sophisticated plan is $(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})K$ (`Notas/L03/L03-08-time-inconsistency.md` §2.2). Use the corrected plan as the no-commitment benchmark in (b). Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **The Value of Commitment:** Consider the three-period example of a player with hyperbolic discounting described in Section 8.3.4 with utility $\ln(x)$ in each of the three periods and with discount factors $\delta = 1$ and $\beta = \frac{1}{2}$. We solved the optimal consumption plan of a sophisticated player 1.
>
> a. Imagine that an external entity can enforce any plan of action that player 1 chooses in $t = 1$ and will prevent player 2 from modifying it. What is the plan that player 1 would choose to enforce?
>
> b. Assume that $K = 90$. Up to how much of his initial budget K will player 1 be willing to pay the external entity in order to enforce the plan you found in (a)?

Referred to in the statement. The three-period example of Section 8.3.4 and its sophisticated solution (Tadelis pp. 166 and 168–169 [PDF p. 185 and 187–188]) (reproduced from the book; the source note above flags the printed $x_1(K) = \frac{2}{3}K$):

> Consider a player with $u(x) = \ln(x)$ who needs to allocate a fixed budget K across three periods. Since he will not waste any of his budget, it follows that $x_1 = K - x_2 - x_3$ [...]
>
> To model this idea consider a different discounting rule known as *hyperbolic discounting*. The player uses the discount rate δ for the future periods as in exponential discounting, but he uses an *additional* discount factor $\beta \in (0, 1)$ to discount all of the future compared to present consumption. Using our three-period example, the player's modified discounted present-value problem is given by
>
> $$\max_{x_2, x_3} v(K - x_2 - x_3, x_2, x_3) = \ln(K - x_2 - x_3) + \beta\delta \ln(x_2) + \beta\delta^2 \ln(x_3). \tag{8.9}$$
>
> [...] Imagine that the player is very well aware of his self-control problem—at period $t = 1$ he discounts between periods $t = 2$ and $t = 3$ at the rate of δ, but he knows that when tomorrow ($t = 2$) comes around, he will discount $t = 3$ at a rate of $\beta\delta$. In other words, the player at $t = 1$ (call him player 1) is aware that he is playing a game with his "future self" at $t = 2$ (call him player 2), who will exhibit different preferences over time. In this case a rational, forward-looking player 1 will solve his problem using backward induction; in other words he will solve for the subgame-perfect equilibrium of the implied dynamic game.
>
> To make things simple, fix $\delta = 1$. Recall from equations (8.4)–(8.6) that a player who uses exponential discounting with $\delta = 1$ will choose to equalize his consumption over time, so that $x_t = \frac{K}{3}$ for all $t \in \{1, 2, 3\}$. Now consider a sophisticated player with hyperbolic discounting with $\delta = 1$ and $\beta = \frac{1}{2}$ (you are asked to solve for the more general problem in exercise 8.15). Looking forward to his future self, he knows that player 2, who is left with a budget of K_2, will solve the following problem (remember that $\delta = 1$ and $\beta = \frac{1}{2}$):
>
> $$\max_{x_2} v_2(x_2, K_2 - x_2) = \ln(x_2) + \tfrac{1}{2}\ln(K_2 - x_2),$$
>
> for which the first-order condition is
>
> $$\frac{dv_2}{dx_2} = \frac{1}{x_2} - \frac{0.5}{K_2 - x_2} = 0,$$

which in turn implies that player 2's best-response function is

$$x_2(K_2) = \tfrac{2}{3}K_2.$$

As a consequence $x_3(K_2) = \frac{1}{3}K_2$. Hence the choices $x_2(K_2)$ and $x_3(K_2)$ made by player 2 are the solution to any of the (infinitely many) subgames that start at the node at which player 2 is left with a budget of K_2. Now moving up to the root of the game, where player 1 decides how much to allocate between his own consumption and that of player 2, he must solve the following problem:

$$\max_{x_1} v_1\left(x_1, \tfrac{2}{3}(K-x_1), \tfrac{1}{3}(K-x_1)\right) = \ln(x_1) + \tfrac{1}{2}\ln\left(\tfrac{2}{3}(K-x_1)\right) + \tfrac{1}{2}\ln\left(\tfrac{1}{3}(K-x_1)\right),$$

for which the first-order condition is

$$\frac{dv_1}{dx_1} = \frac{1}{x_1} - \frac{2}{3}\times\frac{0.5}{K-x_1} - \frac{1}{3}\times\frac{0.5}{K-x_1} = 0,$$

which in turn implies that player 1's best-response function is

$$x_1(K) = \tfrac{2}{3}K.$$

Optional extras.

- Tadelis 8.1 and 8.2 (p. 170): mixed-strategy SPE of the voluntary Battle of the Sexes (read Fig. 8.2, p. 154; the slides' Example 2) and of mutually assured destruction (read §8.3.3, pp. 163–166, Fig. 8.11).

- Tadelis 8.9 (entry deterrence with a fixed cost, p. 172) and 8.14 (hyperbolic discounting with general δ, β, p. 174): harder versions of items 1 and 9.

- Muñoz-Garcia Ch. 4 Ex. 2 (entry-predation, p. 156), Ex. 5–6 (sequential Pareto coordination and chicken, pp. 166, 169), Ex. 13–15 (Stackelberg variants, pp. 189, 192, 195): all with worked answers.

(a) What is asked. The three-period problem of Tadelis 8.15 for a *sophisticated* player 1 ($\ln x$, $\delta = 1$, $\beta = \frac{1}{2}$): (a) the plan she would choose if an outside entity could enforce it and stop player 2 from changing it; (b) for $K = 90$, the largest part of the budget she would pay for that enforcement.

(b) Method. Commitment shrinks the set of strategies to a single path, so player 1 simply maximizes her own objective over plans; without commitment the outcome is the SPE of the game between the two selves, solved by backward induction (Tadelis §8.3.4, pp. 166–169; Lecture 4 on credibility and commitment, slides 19–20). Her willingness to pay is the fee that leaves her indifferent, evaluated with her own period-1 objective $v_1 = \ln x_1 + \frac{1}{2}\ln x_2 + \frac{1}{2}\ln x_3$. This is reading, not in the slides of Lecture 4.

(c) Solution. (a) The enforced plan. With budget K the plan maximizing v_1 is the one found in Tadelis 8.15(a):

$$(x_1, x_2, x_3) = \left(\tfrac{K}{2}, \tfrac{K}{4}, \tfrac{K}{4}\right), \qquad v_1^{\text{commit}}(K) = \ln\tfrac{K}{2} + \tfrac{1}{2}\ln\tfrac{K}{4} + \tfrac{1}{2}\ln\tfrac{K}{4} = \ln\frac{K^2}{8}.$$

No commitment (the benchmark). Player 2's reply to a remaining budget K_2 is $x_2 = \frac{2}{3}K_2$, $x_3 = \frac{1}{3}K_2$ (Tadelis 8.15(b)). Player 1 substitutes it, with $K_2 = K - x_1$:

$$v_1(x_1) = \ln x_1 + \tfrac{1}{2}\ln\big(\tfrac{2}{3}(K - x_1)\big) + \tfrac{1}{2}\ln\big(\tfrac{1}{3}(K - x_1)\big).$$

The constants inside the logarithms drop out of the derivative, so $v_1'(x_1) = \frac{1}{x_1} - \frac{1}{K - x_1} = 0$ and $x_1 = \frac{K}{2}$; the sophisticated plan is

$$\big(\tfrac{K}{2}, \tfrac{K}{3}, \tfrac{K}{6}\big), \qquad v_1^{\mathrm{SPE}}(K) = \ln\tfrac{K}{2} + \tfrac{1}{2}\ln\tfrac{K}{3} + \tfrac{1}{2}\ln\tfrac{K}{6} = \ln\frac{K^2}{2\sqrt{18}} = \ln\frac{K^2}{6\sqrt{2}}.$$

Note that this is not the $x_1 = \frac{2}{3}K$ printed on p. 169: with the objective as printed, the maximum is at $\frac{K}{2}$, as the derivative above shows (the book multiplies each term by the constant inside the logarithm). The comparison uses the corrected plan.

(b) The fee. If she pays c for enforcement, the budget is $B = K - c$ and she gets $v_1^{\mathrm{commit}}(B) = \ln\frac{B^2}{8}$. She is indifferent when

$$\frac{B^2}{8} = \frac{K^2}{6\sqrt{2}} \iff B^2 = \frac{4}{3\sqrt{2}}K^2 = \frac{2\sqrt{2}}{3}K^2 \iff B = \Big(\frac{8}{9}\Big)^{1/4}K,$$

since $\frac{2\sqrt{2}}{3} = \sqrt{\frac{8}{9}}$. Hence the largest fee is

$$c^* = K\Big(1 - \big(\tfrac{8}{9}\big)^{1/4}\Big) \approx 0.0290\,K.$$

For $K = 90$: $B = 90 \cdot 0.97099 \approx 87.39$ and $c^* \approx 2.61$, that is, about 2.9% of the budget.

(d) Interpretation. Commitment is valuable because the selves disagree: player 1 would give period 2 only $\frac{K}{4}$, but player 2 takes $\frac{K}{3}$. The price player 1 is willing to pay is modest here because log utility makes the loss from the revised plan small: the first-period consumption is the same $(\frac{K}{2})$ in both plans, and only the split of the rest changes.

(e) Verification. `check_L04.py` recomputes both objectives with sympy, confirms the plans $(\frac{K}{2}, \frac{K}{4}, \frac{K}{4})$ and $(\frac{K}{2}, \frac{K}{3}, \frac{K}{6})$, solves $v_1^{\mathrm{commit}}(K - c) = v_1^{\mathrm{SPE}}(K)$ for c, asserts $c^* = 90(1 - (\frac{8}{9})^{1/4}) \approx 2.6115$, and checks by a grid search that the commitment plan beats the SPE plan for every fee below c^* and loses above it. Output line: `tadelis-8-16: commit plan (K/2,K/4,K/4); SPE plan (K/2,K/3,K/6); c* = 2.6115 at K = 90`. Tadelis prints no answer. With the plan printed in the book instead of the corrected SPE plan, the fee would come out near 7.6, an overstatement.

(f) Common mistakes.

- Using the book's printed $x_1 = \frac{2}{3}K$ as the no-commitment benchmark: it is not the maximizer of player 1's objective, so the fee is overstated.
- Evaluating the no-commitment plan with player 2's objective: both plans are valued with player 1's period-1 preferences, since she is the one who pays.
- Paying the fee out of the *plan*: the fee reduces the budget before the plan is chosen, so the committed plan is optimal for $K - c$, not K.

Answer. (a) She enforces $(\frac{K}{2}, \frac{K}{4}, \frac{K}{4})$. (b) The fee c satisfies $\ln\frac{(K-c)^2}{8} = \ln\frac{K^2}{6\sqrt{2}}$, so $c^* = K(1 - (\frac{8}{9})^{1/4})$; for $K = 90$, $c^* \approx 2.61$ (about 2.9% of K).